SOFTWARE DESIGN DOCUMENT

NINAAllskySafetyMonitor

NINA safety-monitor plugin using Allsky camera image analysis and WunderStation ISYDNEY478 weather data

Document	Software Design Document
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Revision history

Version	Date	Change
1.0	1 Nov 2026	First issue: use case model, class design, scenarios, traceability matrix.
1.1	1 Nov 2026	Updated to match the slice 1 code: MonitorHost and singleton settings store, IMonitoringCycle and MonitorPorts, settings read every cycle, views as XAML resource dictionaries, source-scanning coverage tests and test fakes, extra settings fields. REQ-06 rationale reworded. See §1.5.

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1 Introduction

1.1 Purpose

This document describes the design of NINAAllskySafetyMonitor, a NINA plugin that reports whether it is safe to image. It analyses the latest image from the roof-mounted Allsky camera for cloud in the region of sky the telescope is pointing at, combines that with rain and wind data from weather station ISYDNEY478, and reports the result to NINA as a safety monitor device.

1.2 Scope

The design covers the use case model, the object-oriented class structure, the main and failure scenarios, and traceability from every requirement in requirements.md v4 to the design elements that realise it. Implementation details below class level are left to the code.

1.3 References

Ref	Document
[R1]	NINAAllskySafetyMonitor Requirements, version 4, 1 November 2026 (requirements.md)
[R2]	N.I.N.A. source code, github.com/isbeorn/nina (commit 717ea13, 23 Oct 2026): IEquipmentProvider, ISafetyMonitor, ITelescopeConsumer, NINA.Astrometry
[R3]	N.I.N.A. plugin template, github.com/isbeorn/nina.plugin.template
[R4]	Weather.com PWS current-observations API (response format confirmed 1 Nov 2026)

1.4 Conventions

- Diagrams use UML notation drawn in the Rational Rose style. Sources are PlantUML text files kept with the code.
- The dotted label at the foot of each class (for example `REQ-23 REQ-26`) lists the requirements the class implements. In code these become `[Requirement("REQ-23", "REQ-26")]` attributes.
- Defaults shown in brackets in the requirements are held in the JSON settings (§2.5); class diagrams show them as initial values.

1.5 Changes in version 1.1

Version 1.1 brings the design into line with the code written for slice 1 (skeleton) and with decisions taken while writing it. The behaviour and requirement allocation are unchanged.

Diagram	Change	Reason
CL-02	HorizontalCoordinate and PixelPoint are record structs.	Small values created in large numbers during star matching; structs avoid heap allocation.
CL-03b	Added SchemaVersion, DirectSourceSettings.Port, WeatherSettings.TimeoutS; SettingsStore gains Path, ToJson, FromJson, Clone; write-then-replace save.	Settings migration, non-standard SSH port, weather timeout (REQ-51), safe editing of a copy and crash-safe saving.
CL-06	SiteLocationResolver takes the fallback as a function; a profile at 0°, 0° counts as unset. OperatingWindow takes the limit per call.	Settings are read live each cycle; NINA profiles default to 0°, 0°.
CL-08	Added IMonitoringCycle and MonitorPorts; MonitorService takes interval and FailSafeGuard takes timeout as functions; MonitoringCycle reads settings each cycle.	Testable with a fake cycle; one object carries the host's adapters; saved settings apply without reconnecting.
CL-10a	Added MonitorHost (single instance). PluginSettingsStore is a single instance holding the current settings. The device delegates to MonitorHost.	NINA creates the plugin, device and dock panel separately; they must share one monitor and one settings copy.

Diagram	Change	Reason
CL-10b	Views are XAML resource dictionaries (StatusTemplates, OptionsTemplates, CalibrationTemplates) matched by key. SettingsViewModel edits a copy, with Save and Revert. Added DelegateCommand. Duplicate Views package removed.	How NINA locates plugin views; edits apply only on Save.
CL-11	Coverage tests scan source files; added RequirementTagScanner, RepositoryPaths, RequirementInfo, CoverageReport and the Fakes package.	The plugin assemblies cannot be loaded on the Mac; fakes make Core testable without NINA.
SD-01, SD-07, SD-10, SD-11	Connect, failure, disconnect and status/settings flows go through MonitorHost; settings read every cycle; Revert added.	Follows from CL-08 and CL-10a/b.
§2.3, CL-06	REQ-06 rationale reworded to match requirements.md.	Research on the AM5N home-referenced Azimuth frame.

2 Architecture

2.1 Principles

The design is object-oriented with single-role, single-responsibility classes, one class per file. All safety logic is in a class library with no NINA references, so every rule can be unit-tested and replayed offline. NINA-specific code is confined to a thin plugin layer that adapts NINA to the library's interfaces (ports).

2.2 Assemblies

Assembly	Role	Depends on
NINAAllskySafetyMonitor.Core	All safety logic: acquisition, weather, geometry, cloud detection, decision, monitoring loop, tuning, logging, settings, traceability attribute.	— (no NINA)
NINAAllskySafetyMonitor.Astrometry	Implements IAstroCalculator over NINA.Astrometry (NOVAS): sun and Moon position, RA/Dec ↔ alt/az.	Core, NINA.Astrometry
NINAAllskySafetyMonitor.Plugin	NINA plugin: equipment provider, safety-monitor device, adapters for mount, site, credentials and logging, MVVM views.	Core, Astrometry, NINA
NINAAllskySafetyMonitor.Harness	Console tool: offline replay of saved frames (REQ-42) and generation of the traceability workbook.	Core, Astrometry
NINAAllskySafetyMonitor.Tests	Unit tests per class and the requirement-coverage tests.	Core

2.3 NINA integration

Confirmed against the NINA source [R2]: the plugin exports an `IEquipmentProvider<ISafetyMonitor>` through MEF. NINA gathers it with `PluginEquipmentProviders<ISafetyMonitor>` and lists the returned device in Equipment → Safety Monitor alongside other drivers. The device implements `ISafetyMonitor` (`IsSafe` plus the `IDevice` members). Connect builds and starts the monitor; Disconnect stops it, so nothing runs outside NINA (REQ-01, REQ-02).

Mount state arrives through `ITelescopeConsumer.UpdateDeviceInfo` and site location from `IProfileService`. The ROI is always computed from RA/Dec. The driver's reported Alt/Az is logged for diagnostics only, because the ASI Mount driver reports Azimuth in the mount's own home-referenced frame (0° at the celestial pole), not the geographic azimuth the ASCOM standard defines (REQ-06).

NINA creates the plugin, the equipment provider (and its device) and the dock panel independently. They share one `MonitorHost` and one `PluginSettingsStore` (single instances), so there is one monitor and one copy of the settings per NINA session. Connect calls `MonitorHost.StartAsync`; Disconnect and plugin teardown call `StopAsync` (CL-10a, SD-01, SD-10).

2.4 Threading and fail-safe behaviour

MonitorService runs the monitoring cycle on a background task, never on NINA's UI thread. Each cycle runs inside FailSafeGuard, which applies a timeout and a catch-all; any exception, hang or undeterminable state yields an Unsafe status and is logged, and nothing is thrown back to NINA (REQ-49, REQ-50). Every network operation has its own timeout (REQ-51). View models marshal snapshots to the UI dispatcher.

2.5 Settings

All configurable values live in one JSON file (the MonitorSettings tree, CL-03b), loaded and saved by SettingsStore, including the lens calibration, horizon mask and clear-sky reference (REQ-55 to REQ-57). The SFTP password, if used, is held in Windows Credential Manager, and secrets are redacted from all logs (REQ-53, REQ-54). The settings file is `%LOCALAPPDATA%\NINAAllskySafetyMonitor\settings.json`, outside the NINA plugin folder, so reinstalling the plugin keeps it. Saves go to a temporary file which then replaces the old one. MonitoringCycle reads the current settings at the start of every cycle, so a connected monitor applies saved changes on its next cycle without reconnecting.

2.6 Traceability process

Every class and every test carries `[Requirement("REQ-xx")]`. A coverage test parses requirements.md, scans the source files for tags (so the Windows-only projects are covered when tests run on the Mac), and fails the build when a requirement has no implementing class or no test, or when a tag names an unknown requirement. The two completeness checks are skipped until the last slice; a coverage report lists the gaps meanwhile. The harness regenerates traceability.xlsx for review. Section 6 gives the design-level matrix.

2.7 Candidate libraries

Need	Candidate	Notes
SFTP with Ed25519 key and host-key pinning	SSH.NET	HostKeyReceived event for fingerprint check
JPEG decode and pixel access	SixLabors.ImageSharp	Fully managed; harness can run without native libraries
Lens-model least-squares fit	MathNet.Numerics	Levenberg–Marquardt
History chart	OxyPlot	Already used by NINA
Star catalogue	Yale Bright Star Catalogue	Embedded resource, about 9,000 stars

To be confirmed during implementation.

3 Use case model

Six actors interact with the monitor: you as the imager, the NINA sequencer, a timer that drives the monitoring cycle, and three external systems (the Allsky Pi or web server, the Weather.com PWS API, and the mount via NINA). UC-A covers operation, UC-B covers set-up and tuning, and UC-Ca/Cb trace each requirement group to a use case.

UC	Use case	Actors	Requirements
UC-01	Connect safety monitor	Imager, NINA	REQ-01–02, REQ-48
UC-02	Disconnect safety monitor	Imager, NINA	REQ-01–02
UC-03	Report safety status	NINA	REQ-03
UC-04	Run monitoring cycle	Timer	
UC-05	Determine operating window	—	REQ-04, REQ-08–10
UC-06	Acquire Allsky image	Allsky Pi / web server	REQ-11–18, REQ-52–53
UC-07	Determine region of interest	Mount (via NINA)	REQ-04–07, REQ-23–27
UC-08	Detect cloud in ROI	—	REQ-28–31
UC-09	Assess weather	Weather.com PWS API	REQ-32–36
UC-10	Decide safety	—	REQ-37–39
UC-11	Override safety manually	Imager	REQ-40
UC-12	Calibrate lens	Imager	REQ-19–21, REQ-47
UC-13	Define horizon mask	Imager	REQ-22, REQ-47
UC-14	View status	Imager	REQ-45
UC-15	Edit settings	Imager	REQ-46, REQ-53, REQ-55–57
UC-16	Record tuning data	Timer	REQ-41, REQ-43
UC-17	Replay frames offline	Imager	REQ-42
UC-18	Log decision	—	REQ-44, REQ-54
UC-19	Handle monitor failure	—	REQ-49–51

Figure 3-1 UC-A Operation — monitoring and reporting to NINA

The timer drives UC-04, which includes the operating window, image acquisition, ROI, cloud detection, weather and decision use cases. UC-10 includes reporting to NINA and logging. Manual override and failure handling extend the flow.

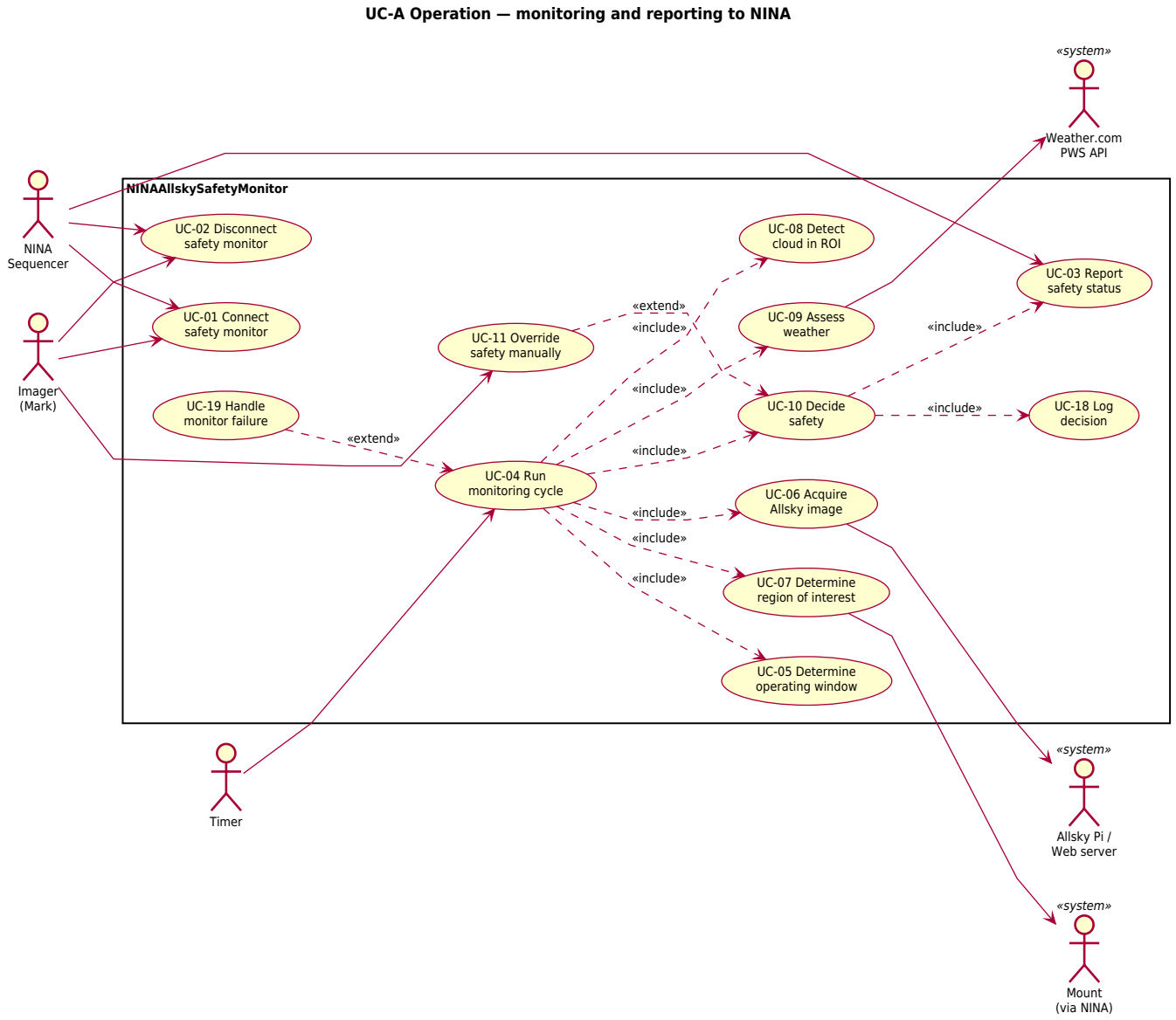


Figure 3-2 UC-B Set-up, review and tuning

Calibrating the lens and drawing the horizon mask both use a live image. Tuning-data recording extends the monitoring cycle when tuning mode is on; offline replay reuses cloud detection.

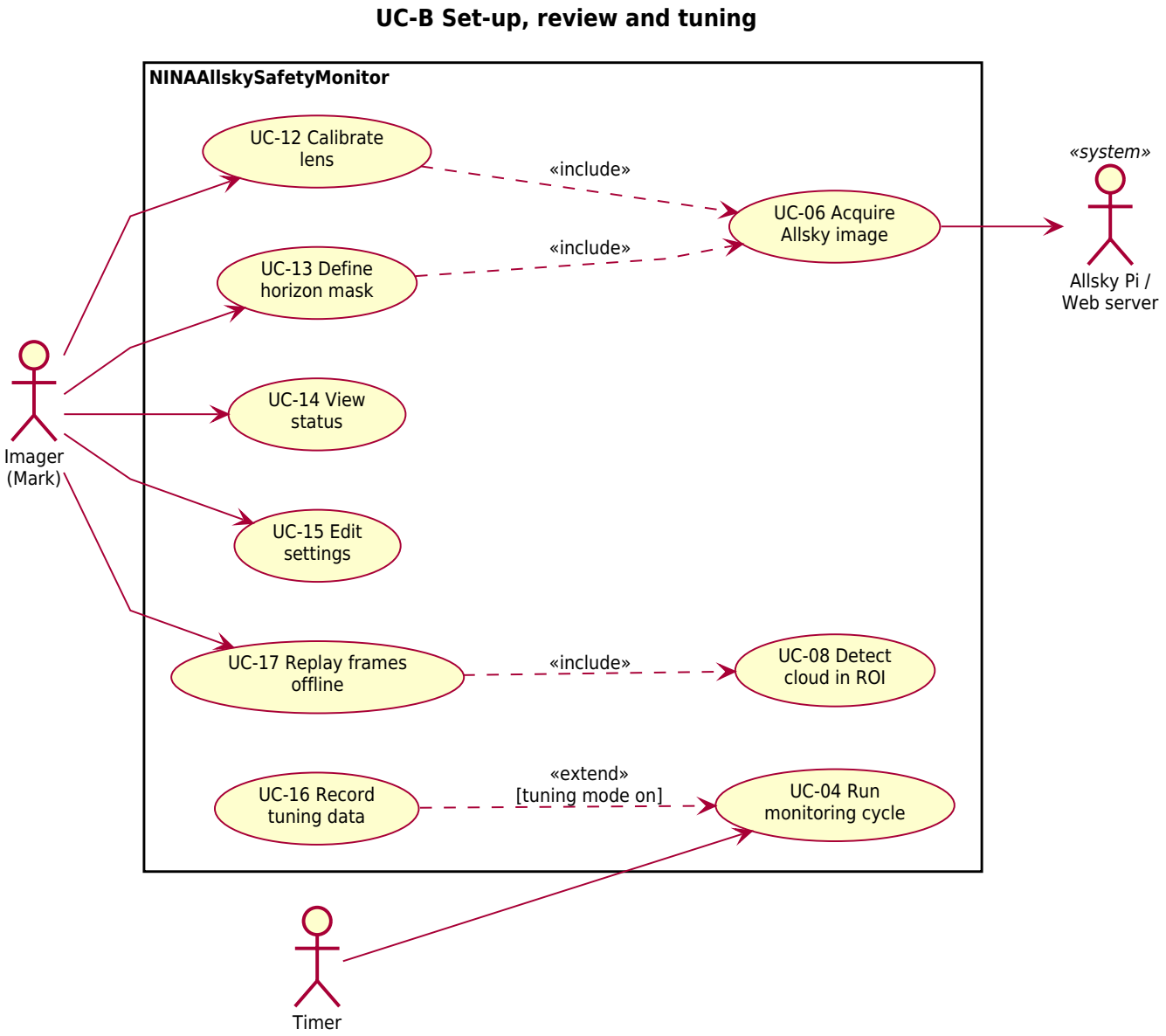


Figure 3-3 UC-Ca Requirements §1–§8 → use cases

Functional requirement groups for NINA integration, site and mount, operating window, acquisition, calibration, ROI, cloud detection and weather.

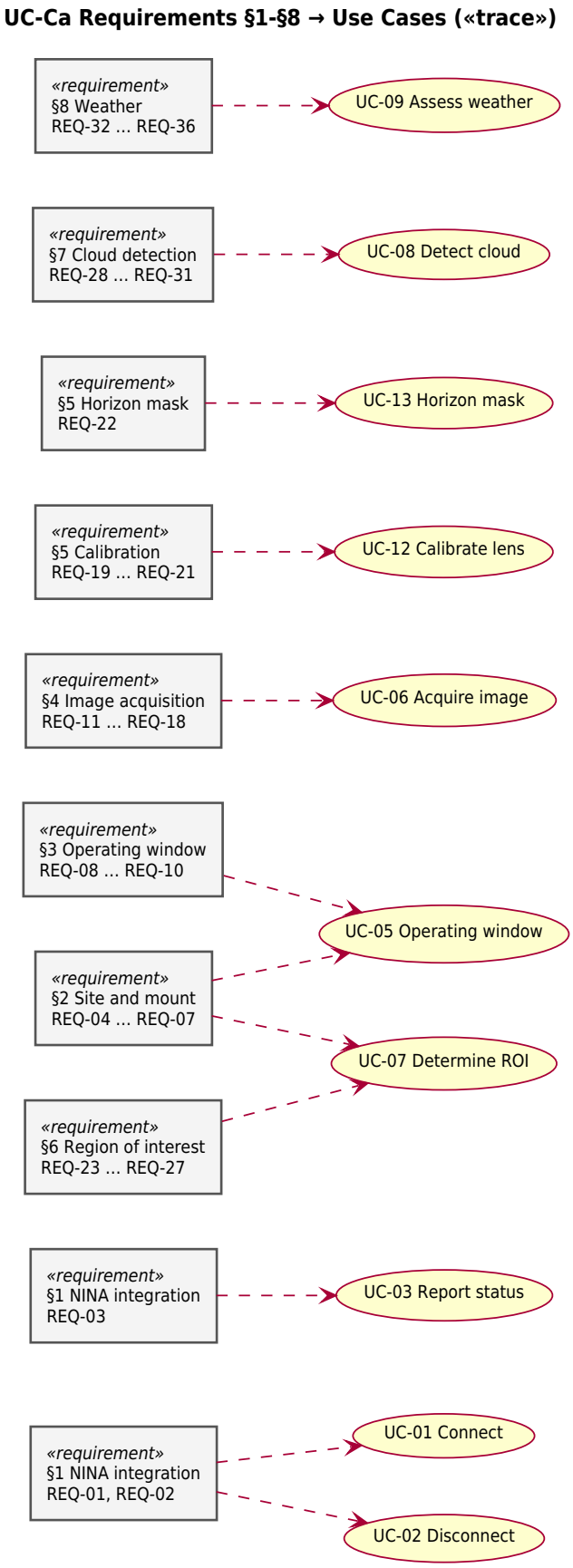
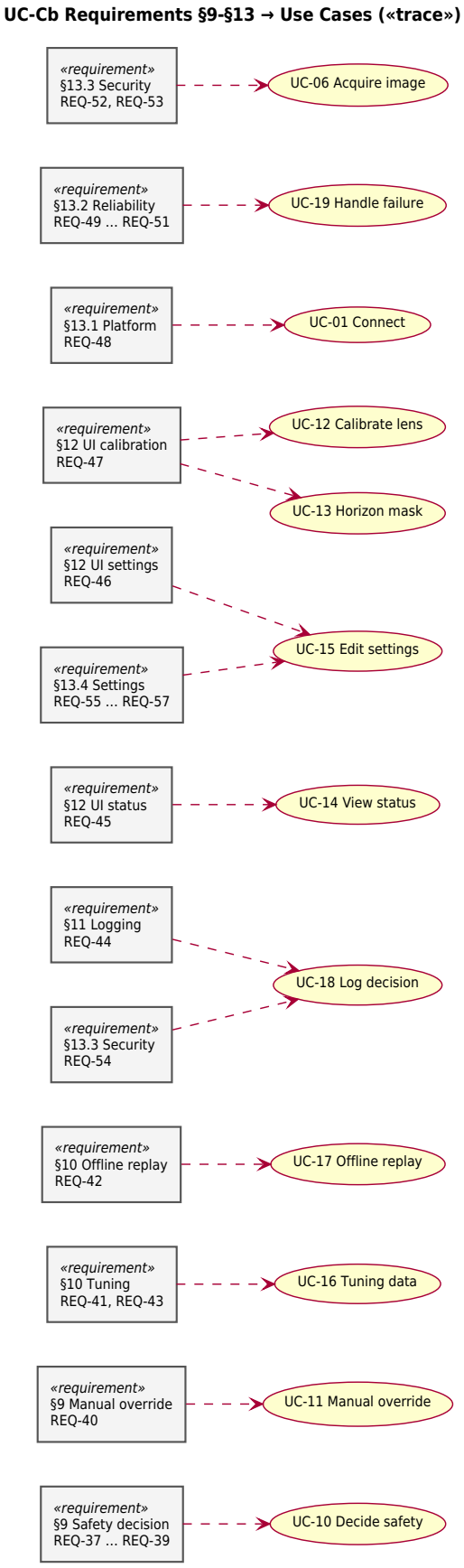


Figure 3-4 UC-Cb Requirements §9–§13 → use cases

Safety decision, tuning, logging, user interface and the non-functional groups.



4 Class design

Thirteen class diagrams describe the design package by package. Table 4-1 lists every class that implements a requirement, with its diagram.

Table 4-1 Classes and the requirements they implement

Diagram	Class	Requirements
CL02	CloudAssessment	REQ-28
CL02	MonitorSnapshot	REQ-41, REQ-45
CL02	MountState	REQ-05–07
CL02	RegionOfInterest	REQ-23, REQ-26
CL02	SafetyConditions	REQ-37
CL02	SafetyStatus	REQ-03, REQ-39–40, REQ-43
CL02	SiteLocation	REQ-04
CL02	SkyImage	REQ-11, REQ-15
CL02	WeatherObservation	REQ-33
CL03a	IAstroCalculator	REQ-08, REQ-24, REQ-31
CL03a	ICredentialStore	REQ-53
CL03a	IMountStateSource	REQ-05
CL03a	ISiteLocationSource	REQ-04
CL03b	CalibrationData	REQ-57
CL03b	MonitorSettings	REQ-55
CL03b	SettingsStore	REQ-55–56
CL04	HostKeyVerifier	REQ-52
CL04	ImageAcquirer	REQ-14, REQ-17, REQ-51
CL04	ImageFreshnessEvaluator	REQ-15, REQ-18
CL04	ImageSourceFactory	REQ-11
CL04	JpegIntegrityChecker	REQ-17
CL04	SftpConnector	REQ-12–13
CL04	SftpImageSource	REQ-11, REQ-15–16
CL04	WebImageSource	REQ-11, REQ-15
CL05	PwsObservationParser	REQ-33
CL05	PwsWeatherClient	REQ-32, REQ-51
CL05	WeatherEvaluator	REQ-34–36
CL05	WeatherPoller	REQ-32
CL06	HorizonMask	REQ-22, REQ-25
CL06	HorizonMaskDefinition	REQ-22
CL06	LensCalibration	REQ-19
CL06	LensProjection	REQ-19, REQ-24
CL06	MountStateInterpreter	REQ-07, REQ-26
CL06	OperatingWindow	REQ-08–10
CL06	RoiCalculator	REQ-06, REQ-23–24, REQ-26
CL06	RoiMaskBuilder	REQ-23, REQ-25
CL06	SiteLocationResolver	REQ-04
CL07	BrightnessTest	REQ-30
CL07	ClearSkyReference	REQ-30, REQ-57

Diagram	Class	Requirements
CL07	ClearSkyReferenceBuilder	REQ-30
CL07	CloudDetector	REQ-28
CL07	ExpectedStarPredictor	REQ-29, REQ-31
CL07	MoonBrightnessAdjuster	REQ-31
CL07	StarCatalogue	REQ-29
CL07	StarFractionTest	REQ-29
CL08	FailSafeGuard	REQ-49–51
CL08	ManualOverride	REQ-40
CL08	MonitorService	REQ-02, REQ-49
CL08	MonitoringCycle	REQ-10, REQ-37
CL08	SafetyConditionsBuilder	REQ-37
CL08	SafetyDecisionEngine	REQ-03, REQ-37, REQ-43
CL08	SafetyHysteresis	REQ-38–39
CL08	SlewHoldPolicy	REQ-27
CL08	StatusHistory	REQ-45
CL09	CalibrationOverlayRenderer	REQ-21
CL09	DecisionLogger	REQ-06, REQ-44
CL09	LensCalibrationSession	REQ-19–20
CL09	LensModelFitter	REQ-19–20
CL09	MetricsCsvWriter	REQ-41
CL09	ReplayImageSource	REQ-42
CL09	RollingFileLogSink	REQ-44
CL09	SecretRedactor	REQ-54
CL09	StarIdentifier	REQ-20
CL09	TuningRecorder	REQ-41
CL09	TuningStateEvaluator	REQ-43
CL10a	AllskySafetyMonitorDevice	REQ-01–03, REQ-49
CL10a	AllskySafetyMonitorProvider	REQ-01
CL10a	AllskySafetyPlugin	REQ-46, REQ-48
CL10a	MonitorHost	REQ-02–03
CL10a	NinaMountStateSource	REQ-05, REQ-07
CL10a	NinaSiteLocationSource	REQ-04
CL10a	NovasAstroCalculator	REQ-08, REQ-24, REQ-31
CL10a	PluginSettingsStore	REQ-55–56
CL10a	WindowsCredentialStore	REQ-53
CL10b	CalibrationViewModel	REQ-19, REQ-21–22, REQ-47
CL10b	HistoryChartViewModel	REQ-45
CL10b	MetricsViewModel	REQ-45
CL10b	OverlayImageComposer	REQ-45
CL10b	SettingsViewModel	REQ-46, REQ-55
CL10b	StatusViewModel	REQ-40, REQ-43, REQ-45
CL11	FixedMountStateSource	REQ-42
CL11	ReplayClock	REQ-42
CL11	ReplayCommand	REQ-42

Figure 4-1 CL-01 Assemblies and packages

Core holds all safety logic and has no NINA references. Domain types are used by every package; those dependencies are omitted for clarity.

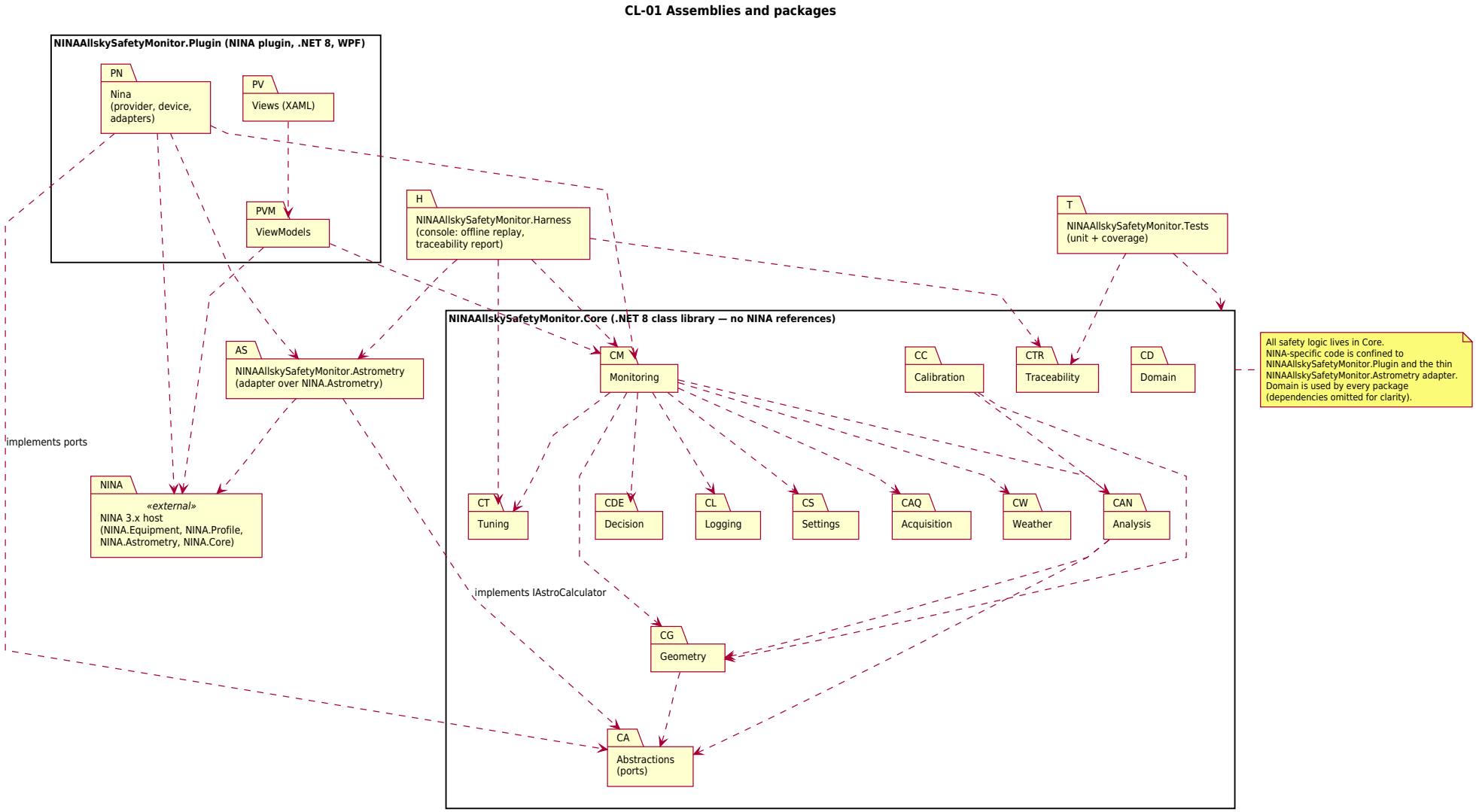


Figure 4-2 CL-02 Domain

Immutable value objects passed between components: images, mount state, ROI, assessments, conditions, safety status and the snapshot used by the UI and tuning.

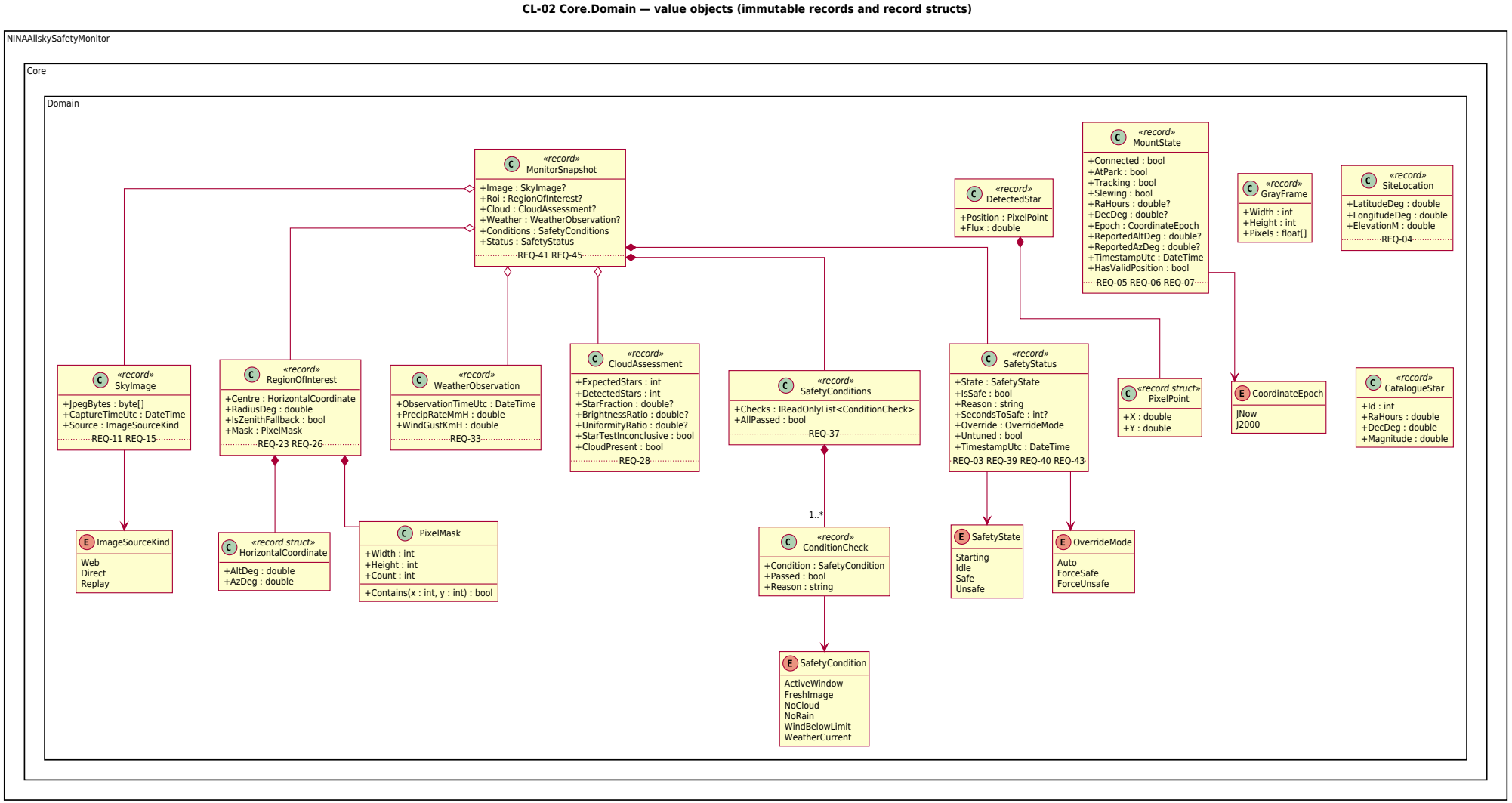


Figure 4-3 CL-03a Ports

Interfaces Core needs from its host. The plugin implements them over NINA; the harness implements them for offline replay.

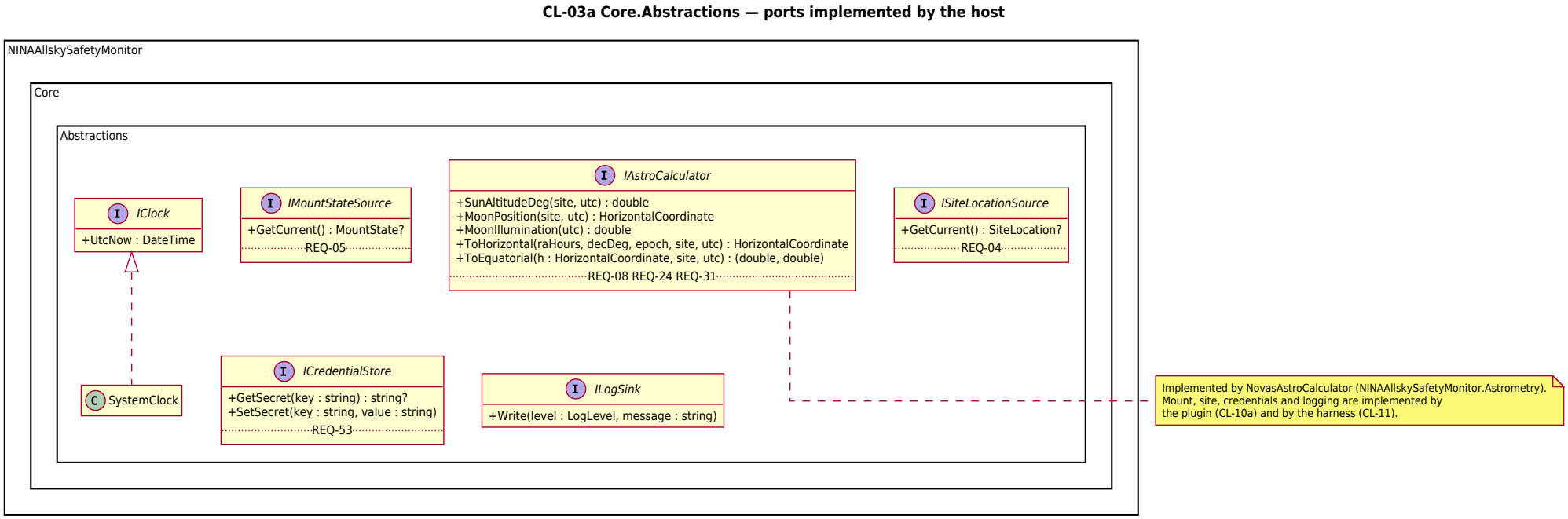


Figure 4-4 CL-03b Settings

The JSON settings tree with its defaults. Initial values match the bracketed defaults in the requirements.

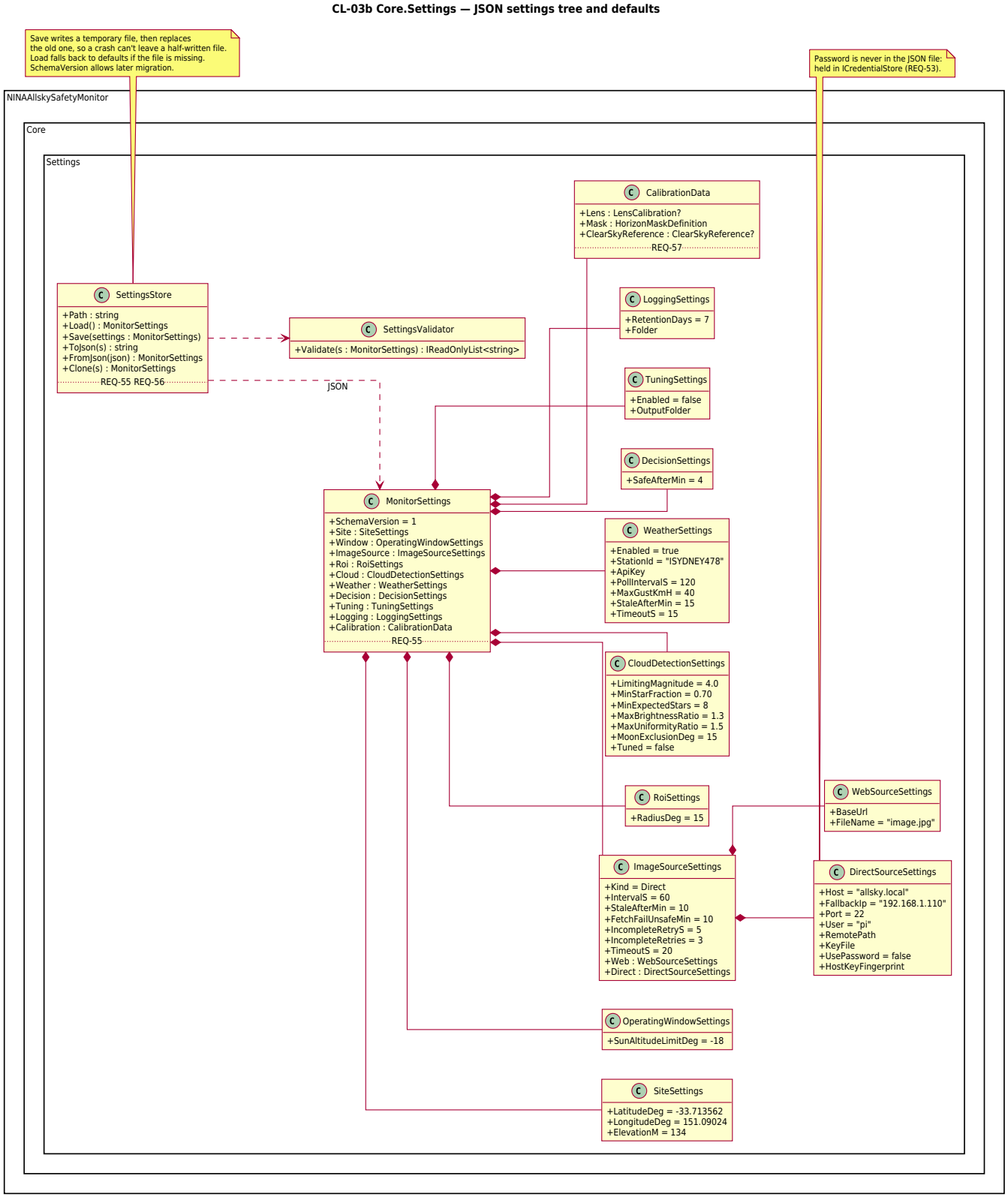


Figure 4-5 CL-04 Image acquisition

Web and SFTP sources behind one interface, host-key pinning, the incomplete-JPEG retry and the freshness rule.

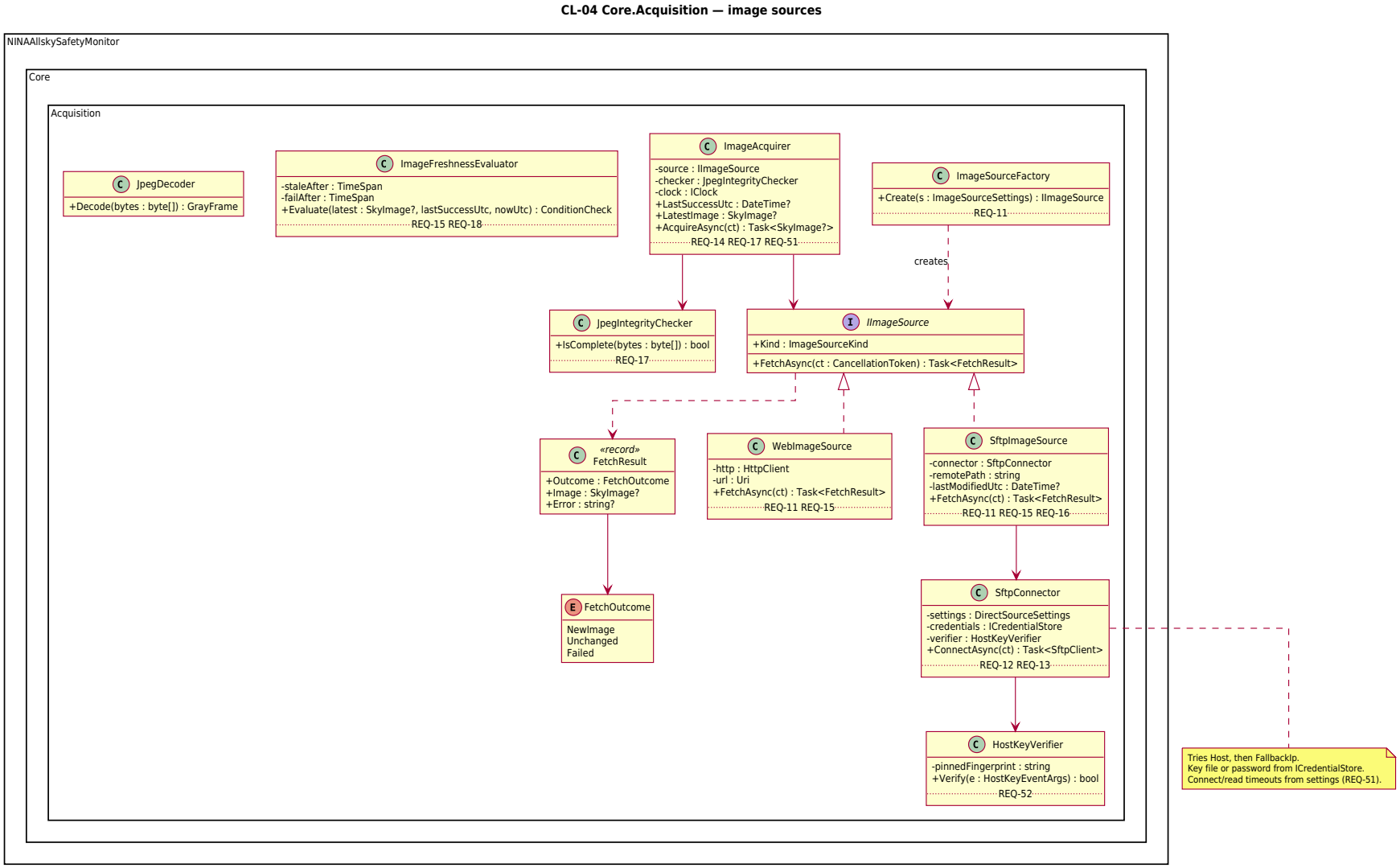


Figure 4-6 CL-05 Weather

PWS client and parser, a 2-minute poller independent of the image cycle, and the evaluator producing the rain, wind and data-age checks.

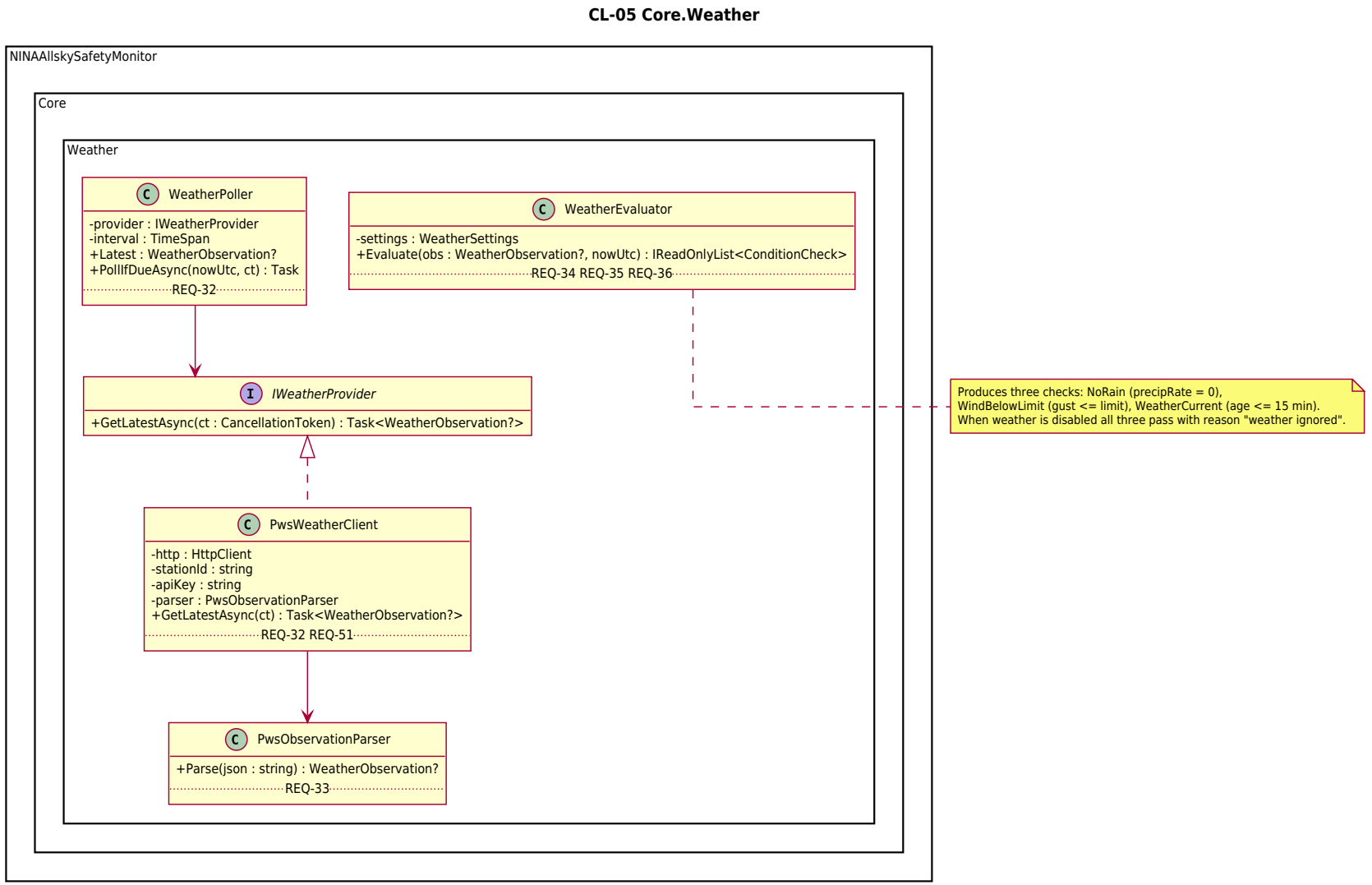


Figure 4-7 CL-06 Geometry

Site resolution, operating window, fisheye lens projection, horizon mask and the ROI built from the mount’s RA/Dec.

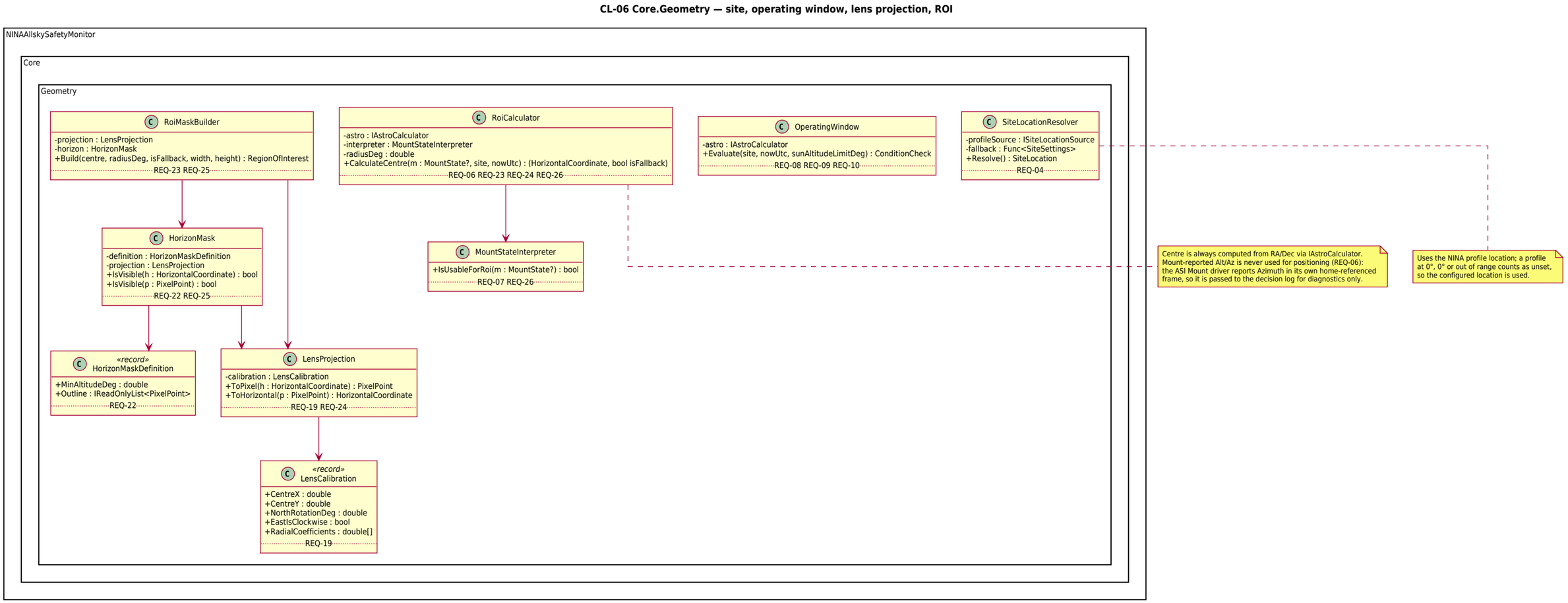


Figure 4-8 CL-07 Cloud detection

Catalogue star prediction, detection and matching, the star-fraction and brightness tests and the Moon adjustment, combined by CloudDetector.

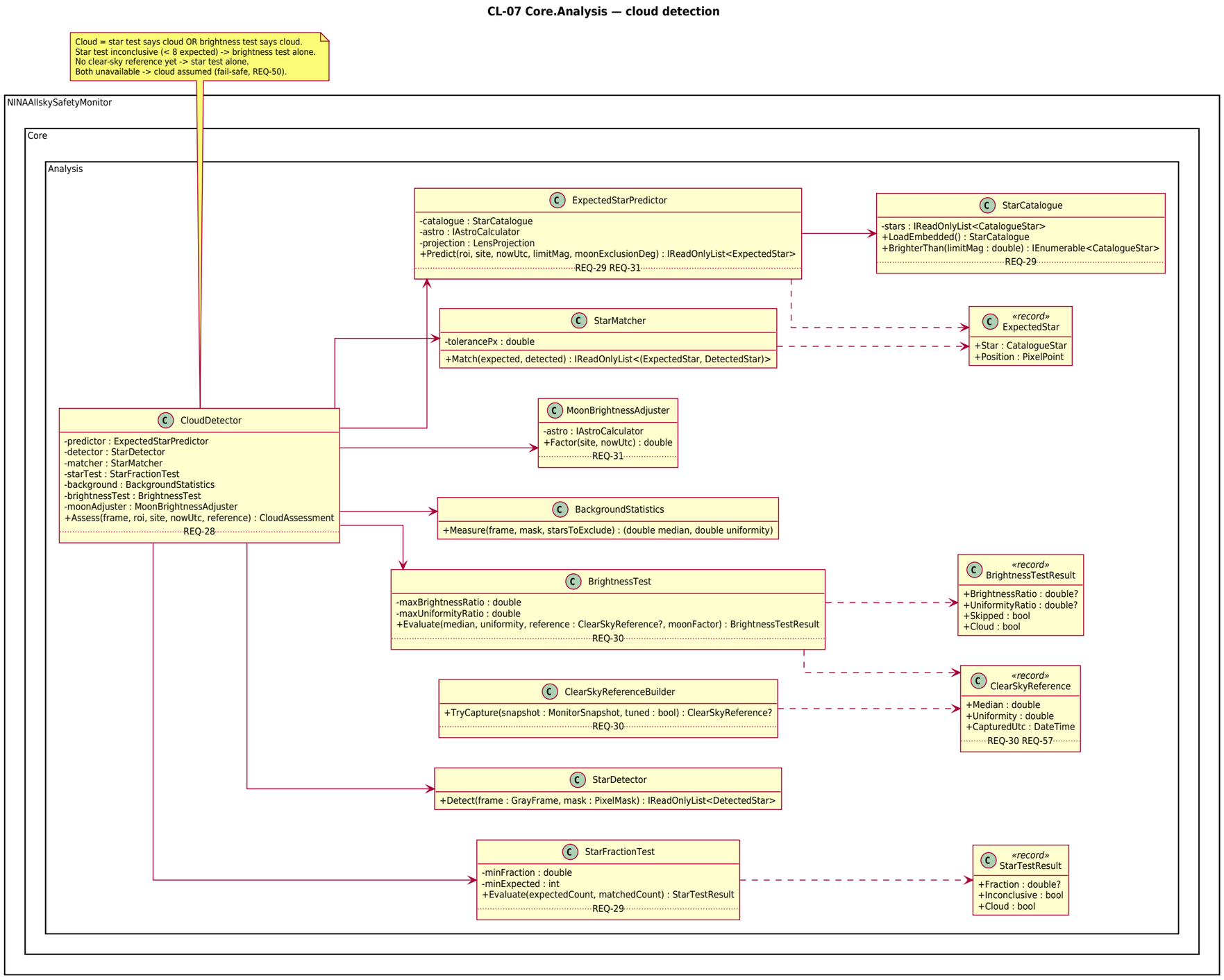


Figure 4-9 CL-08 Decision and monitoring

Conditions pass through hysteresis, slew hold and override to give the status. MonitorService runs each cycle inside FailSafeGuard on a background task.

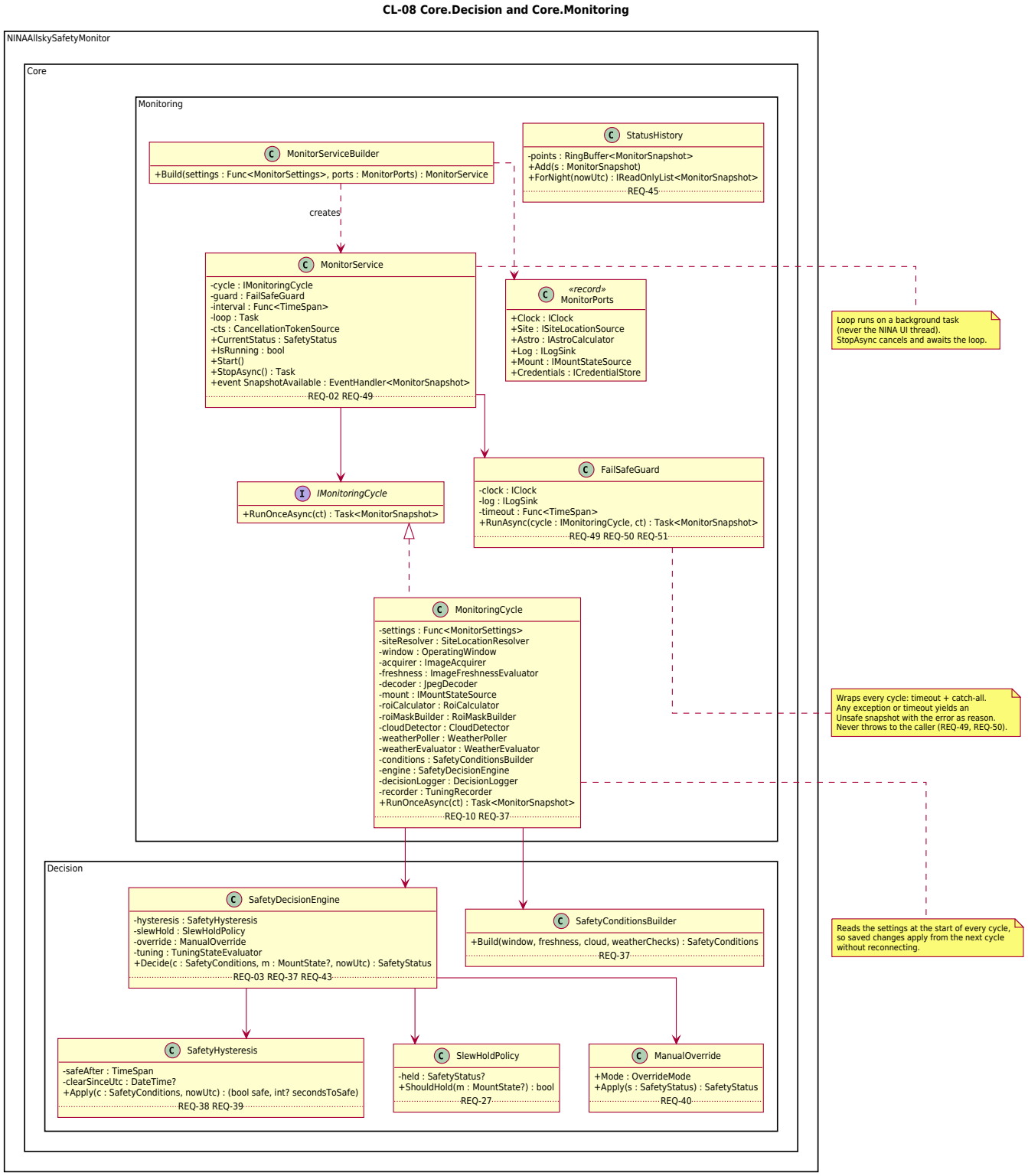


Figure 4-10 CL-09 Calibration, tuning, logging and traceability

Lens calibration from identified stars, the tuning recorder and replay source, the redacting decision log and the Requirement attribute.

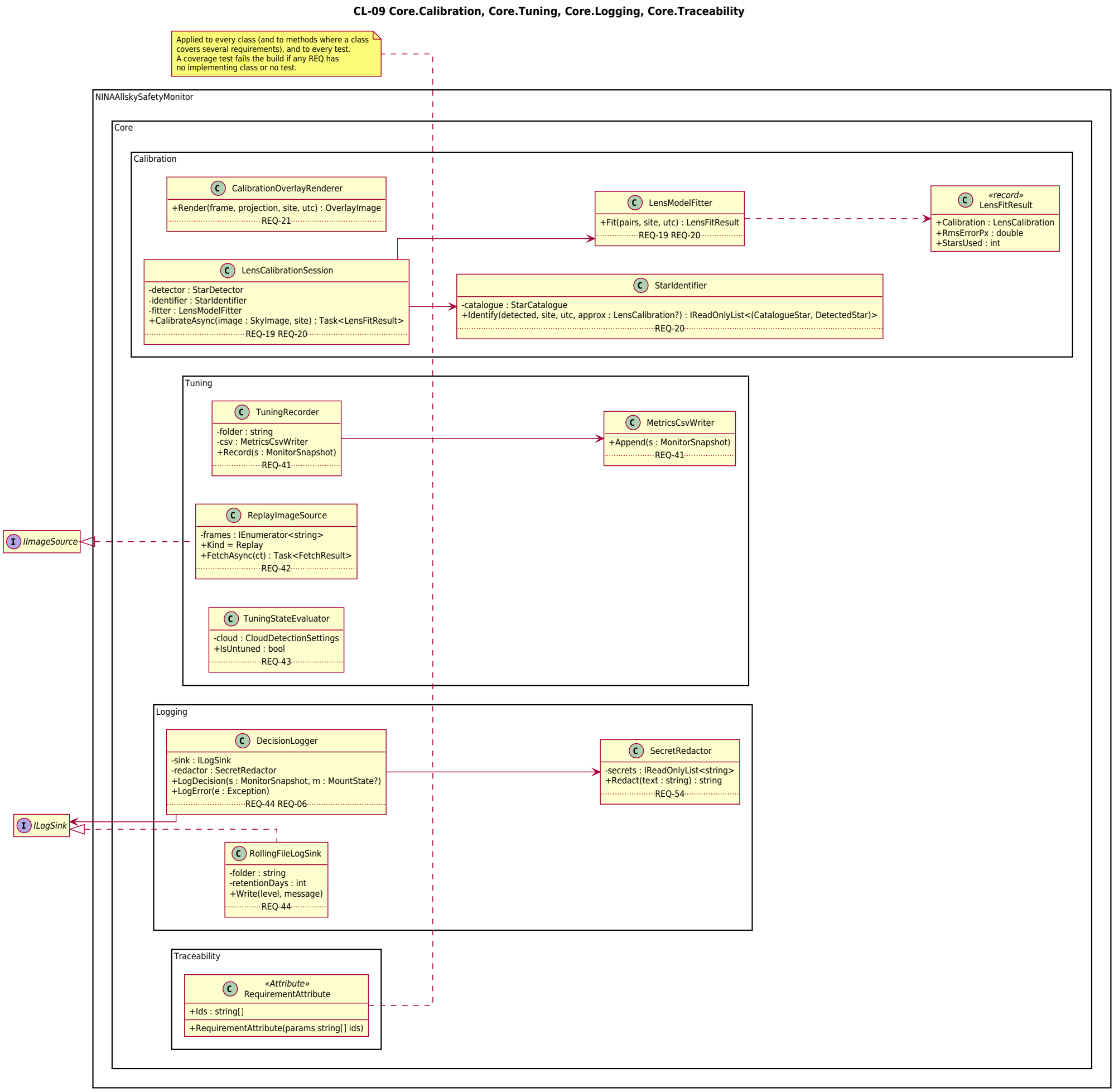


Figure 4-11 CL-10a Plugin: NINA integration

Equipment provider, safety-monitor device, the shared MonitorHost and settings store, and the NINA adapters that implement Core’s ports.

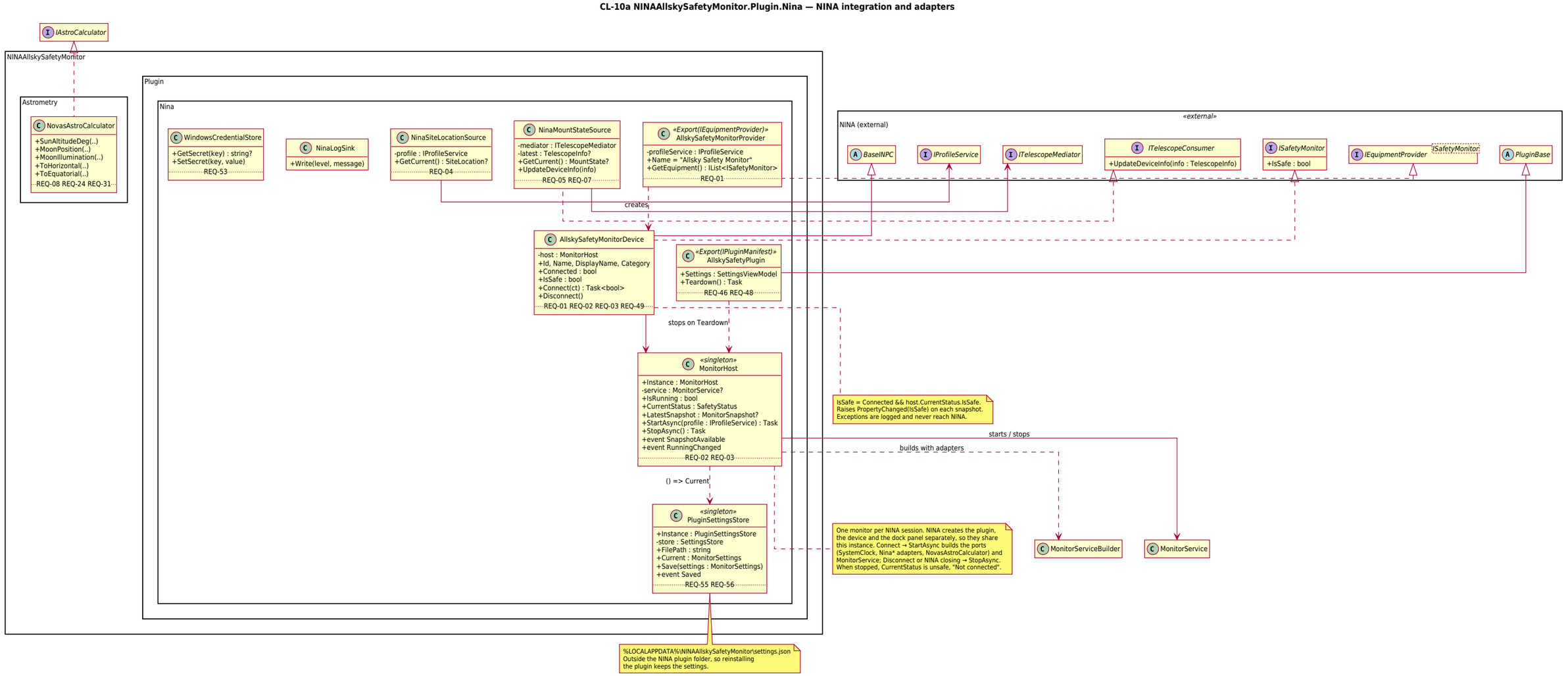


Figure 4-12 CL-10b Plugin: view models and views

MVVM layer inside NINA: the status dock panel, settings page and calibration view.

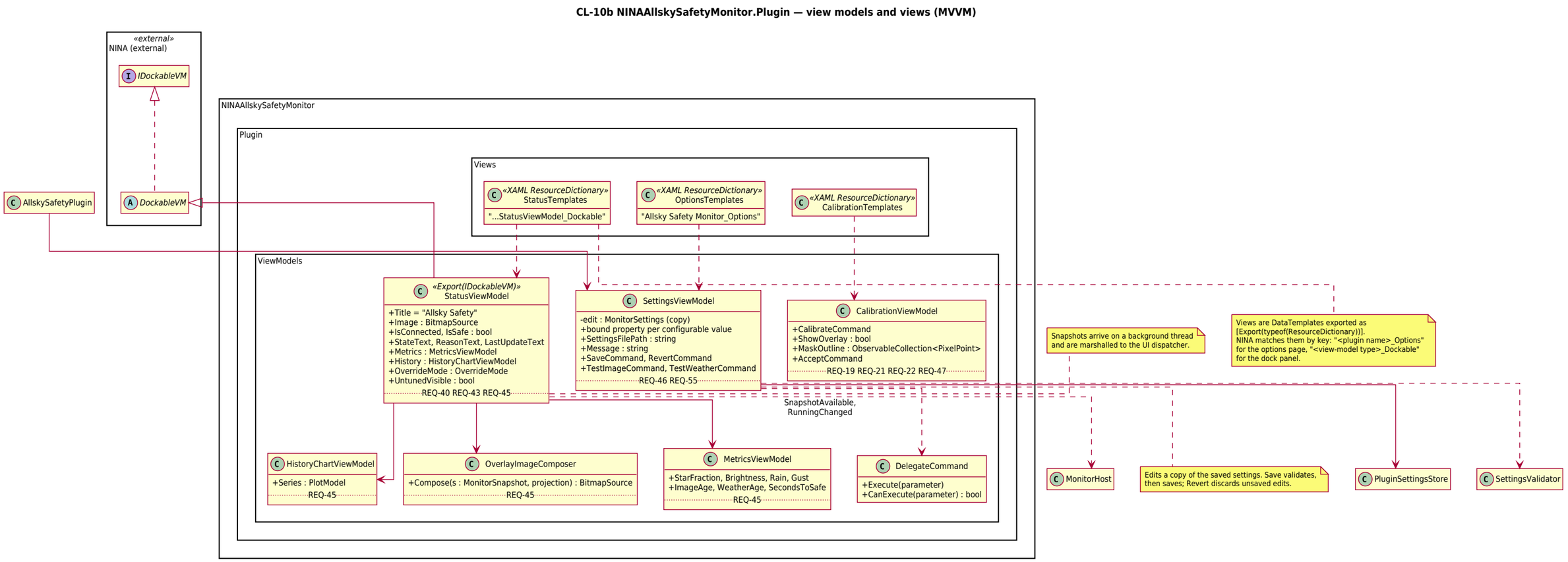
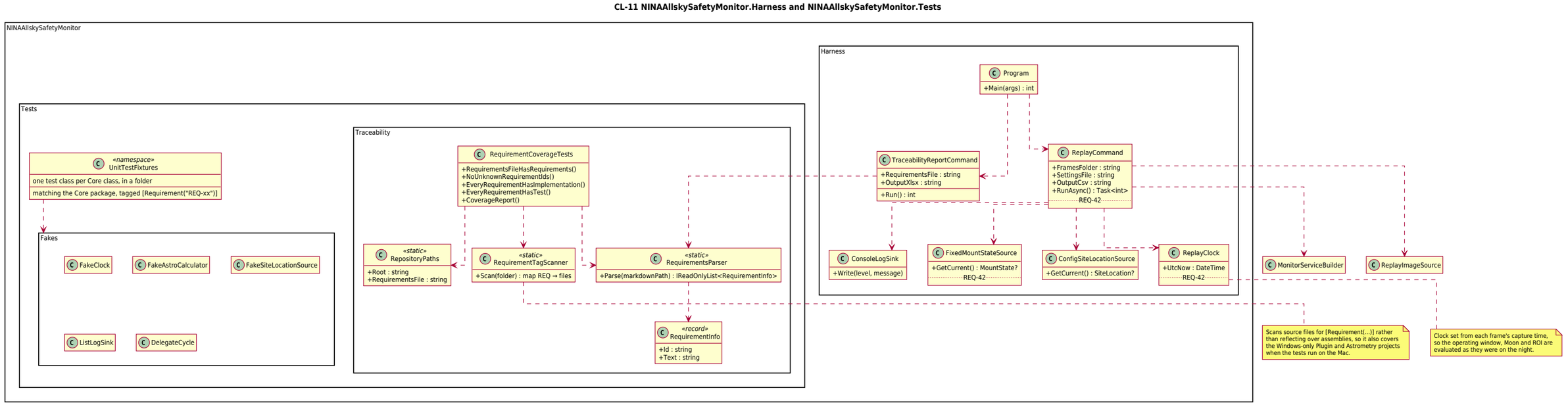


Figure 4-13 CL-11 Harness and tests

Offline replay and traceability-report commands, and the coverage tests.



5 Scenarios

Eleven sequence diagrams cover the main and failure paths. Each title names the use cases and requirements it exercises.

Scenario	Title	Use cases
SD-01	Connect safety monitor	UC-01
SD-02	Monitoring cycle, normal path	UC-04 … UC-10, UC-18
SD-03	Detect cloud in ROI	UC-08
SD-04	Cloud arrives and clears	UC-10, UC-03
SD-05	Slewing, parked or disconnected	UC-07
SD-06	Acquire image over SFTP	UC-06
SD-07	Monitor failure and idle window	UC-19, UC-05
SD-08	Calibrate lens and define horizon mask	UC-12, UC-13
SD-09	Replay saved frames offline	UC-17
SD-10	Manual override and disconnect	UC-11, UC-02
SD-11	View status and edit settings	UC-14, UC-15

Figure 5-1 SD-01 Connect safety monitor

NINA lists the device from the provider at start-up. Connect asks MonitorHost to start: it registers for mount updates, builds the ports and the monitor and starts its background loop. Each cycle reads the current settings. IsSafe stays false until a safe decision.

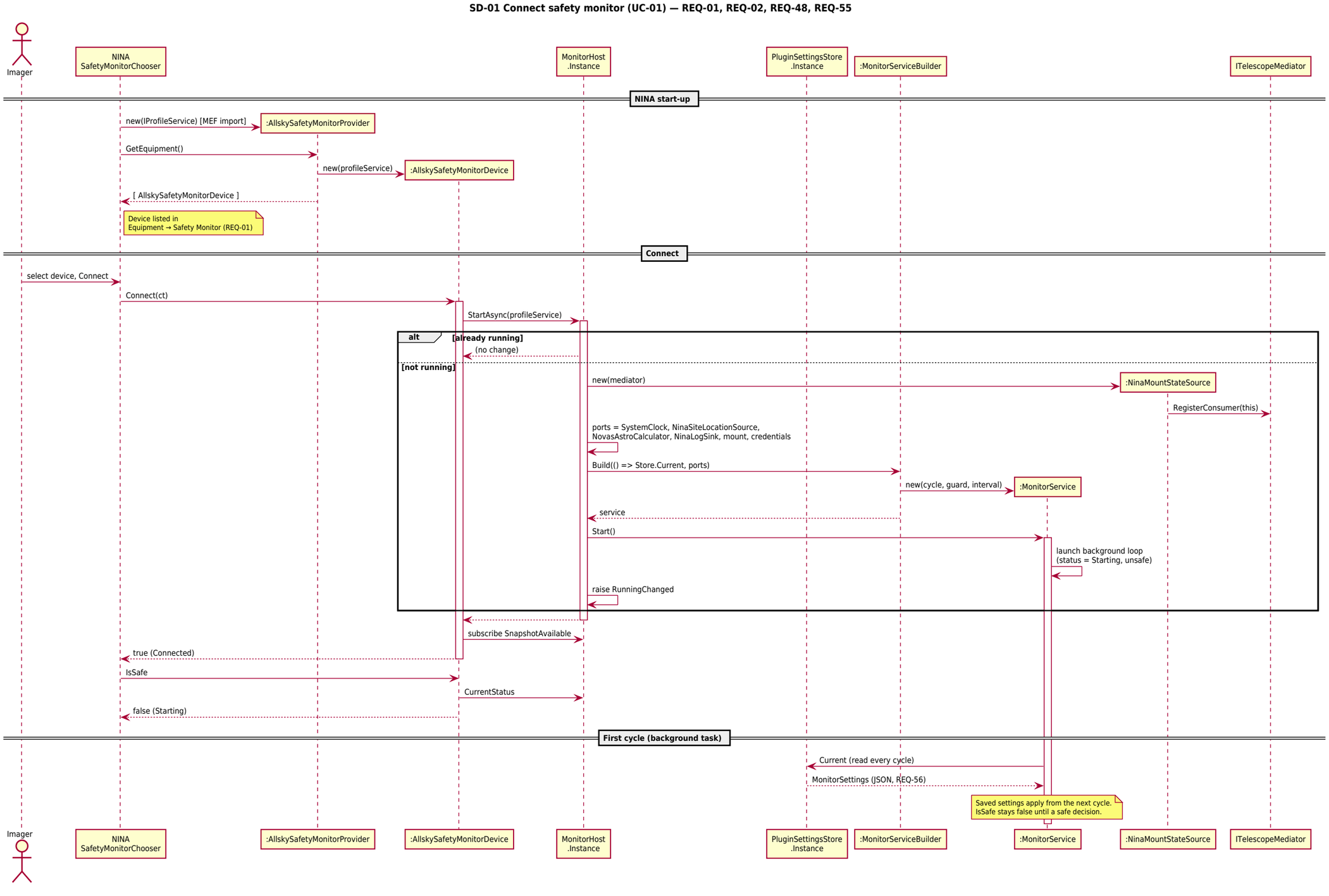


Figure 5-2 SD-02 Monitoring cycle, normal path

One cycle from site and operating window through image, ROI, cloud and weather to the decision, log and optional tuning record.

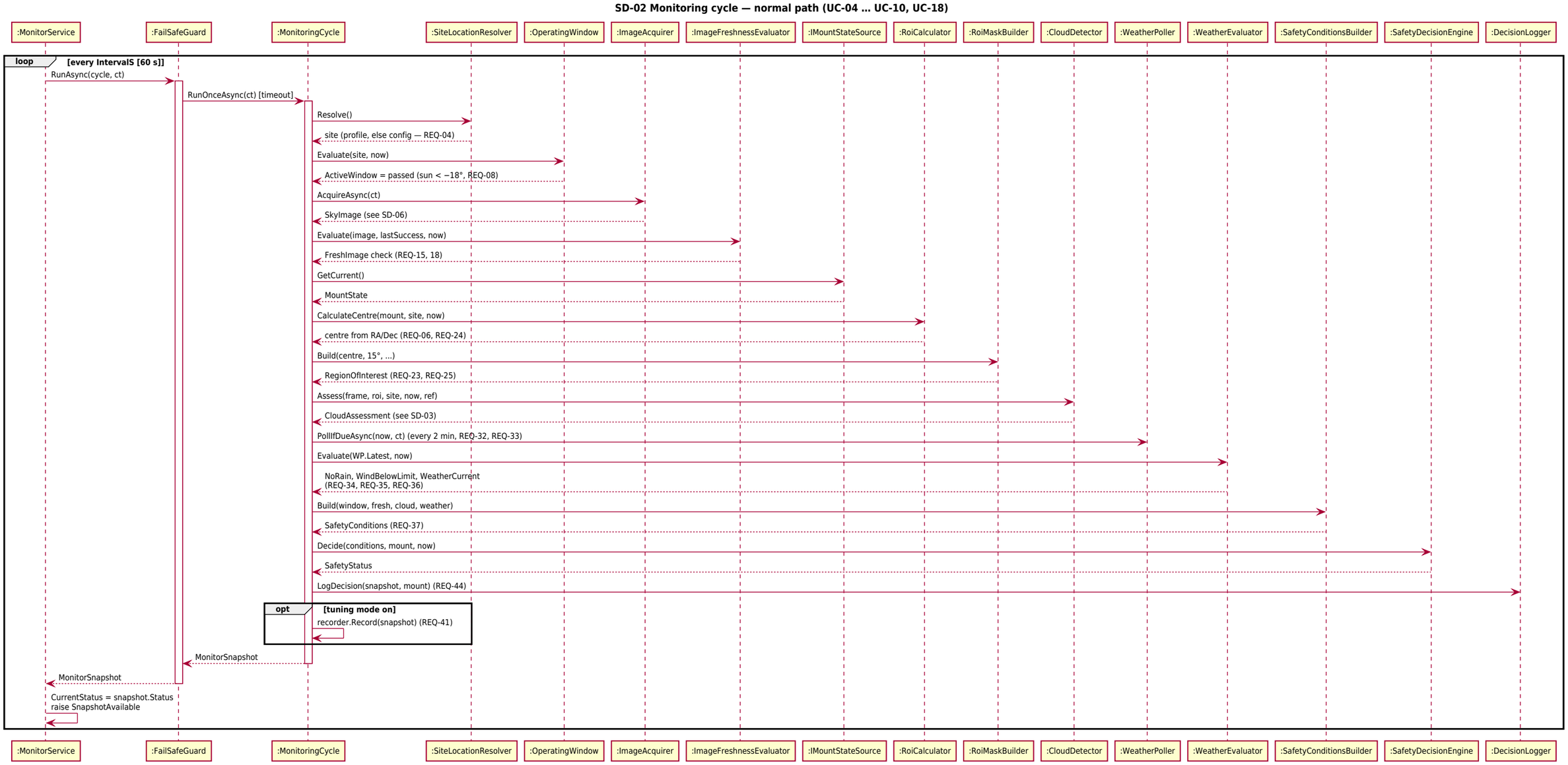


Figure 5-3 SD-03 Detect cloud in ROI

Expected versus detected catalogue stars, the inconclusive case below 8 expected stars, the brightness test against the clear-sky reference, and the fail-safe combination.

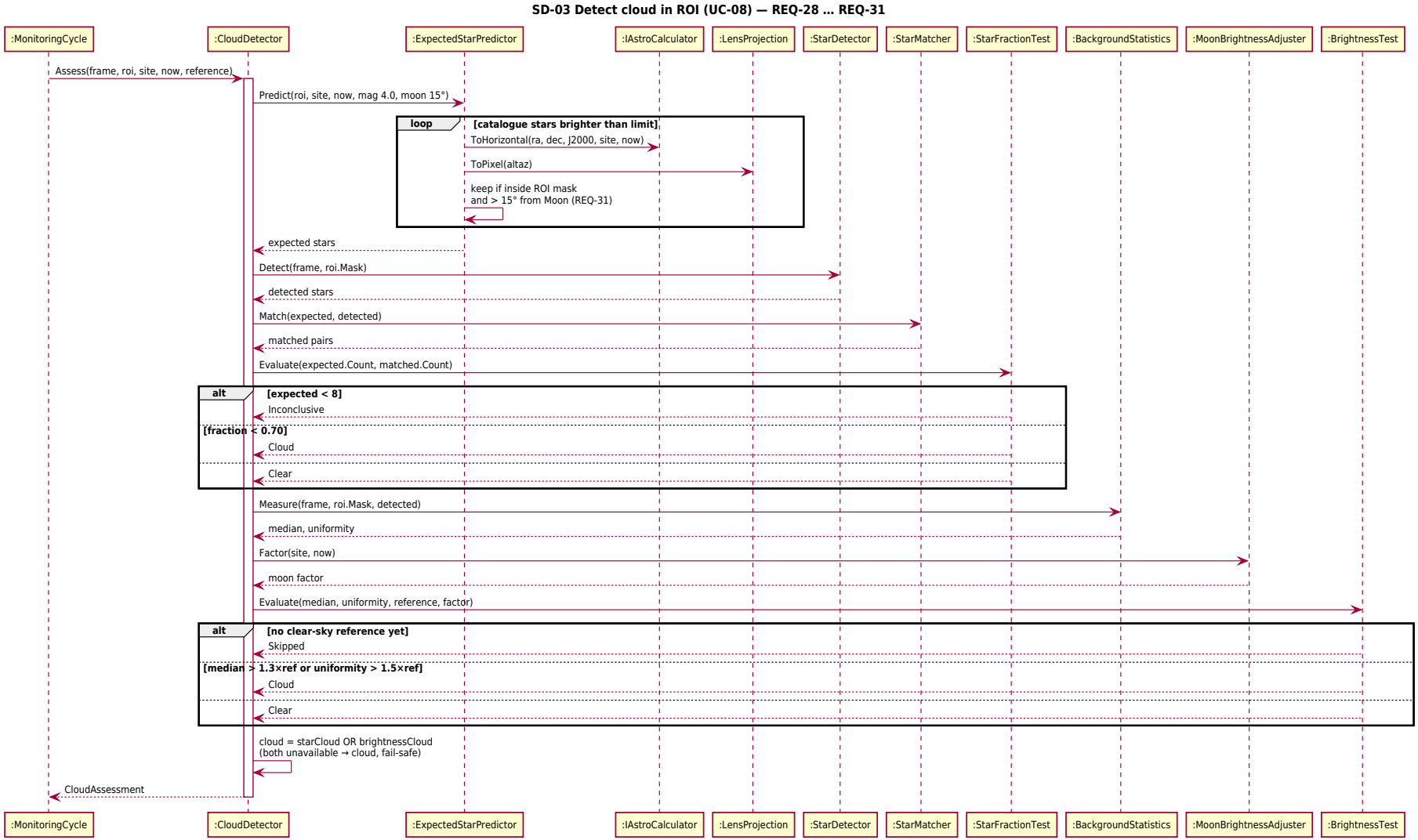


Figure 5-4 SD-04 Cloud arrives and clears

Unsafe is reported at once; safe only after four continuous clear minutes, with the countdown carried in the reason text.

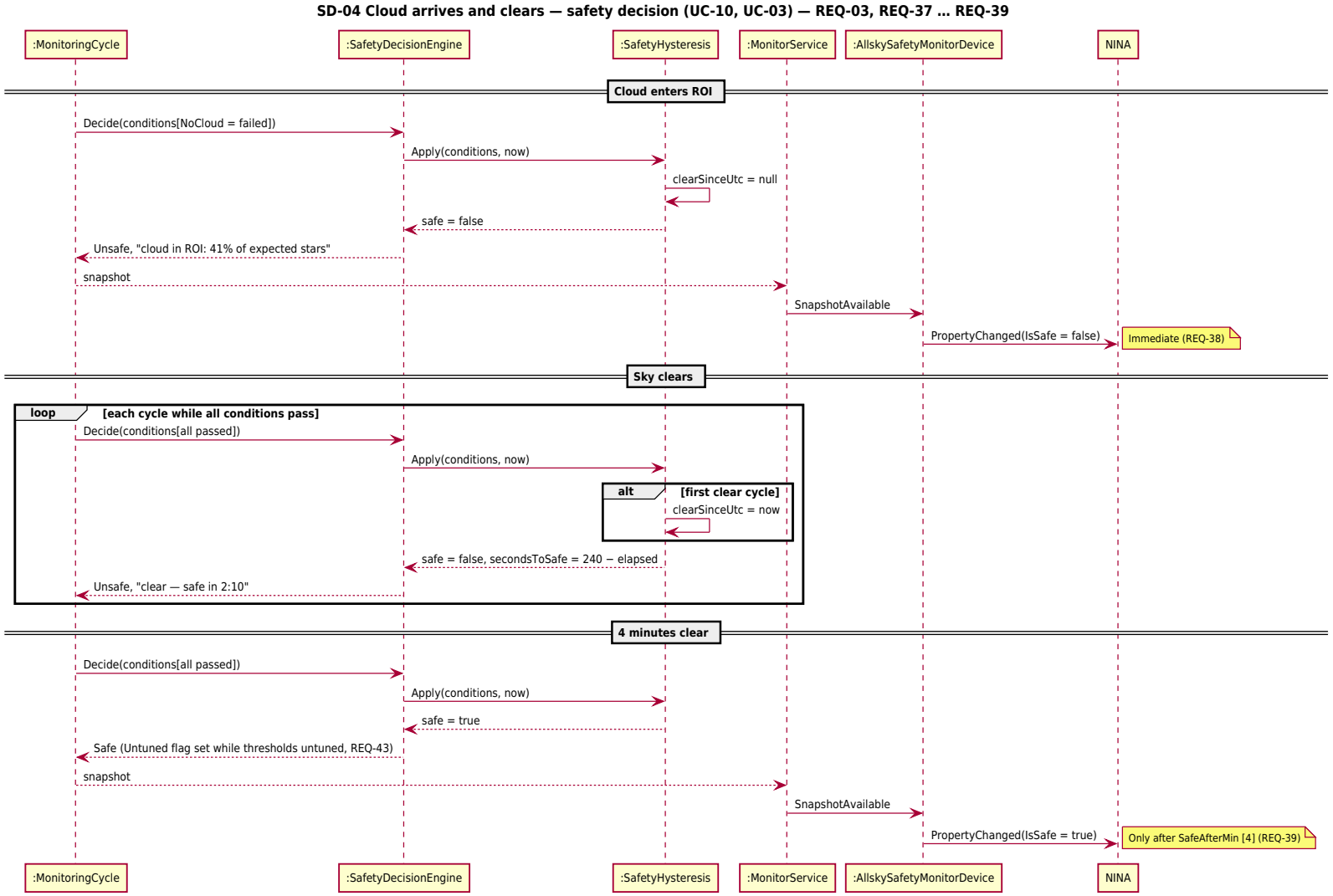


Figure 5-5 SD-05 Slewing, parked or disconnected

The state is held while slewing and the ROI moves after the slew. A parked, non-tracking or disconnected mount gives a zenith ROI.

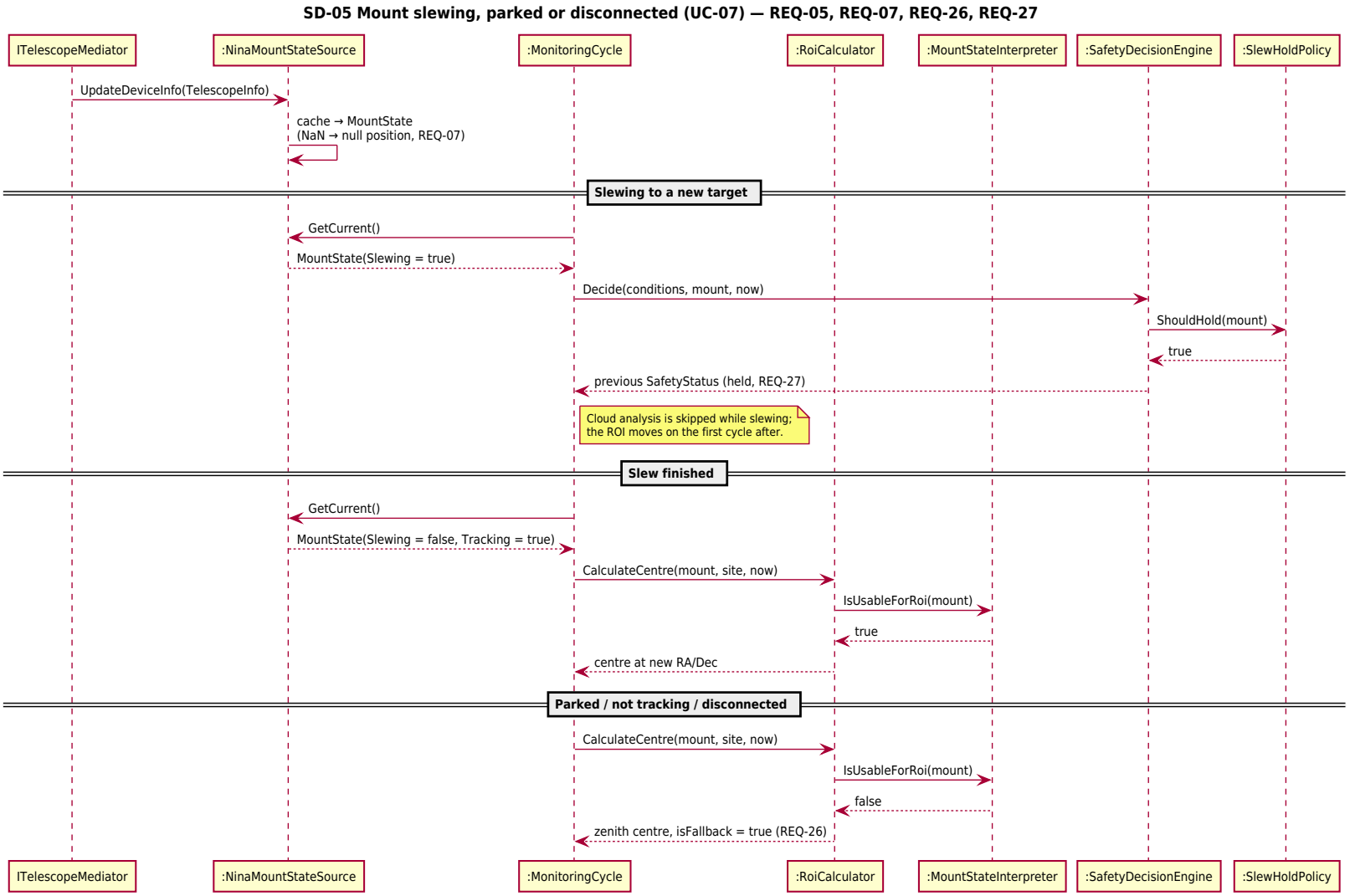


Figure 5-6 SD-06 Acquire image over SFTP

Fallback IP, host-key verification, stat before download, unchanged-image skip and the incomplete-JPEG retry.

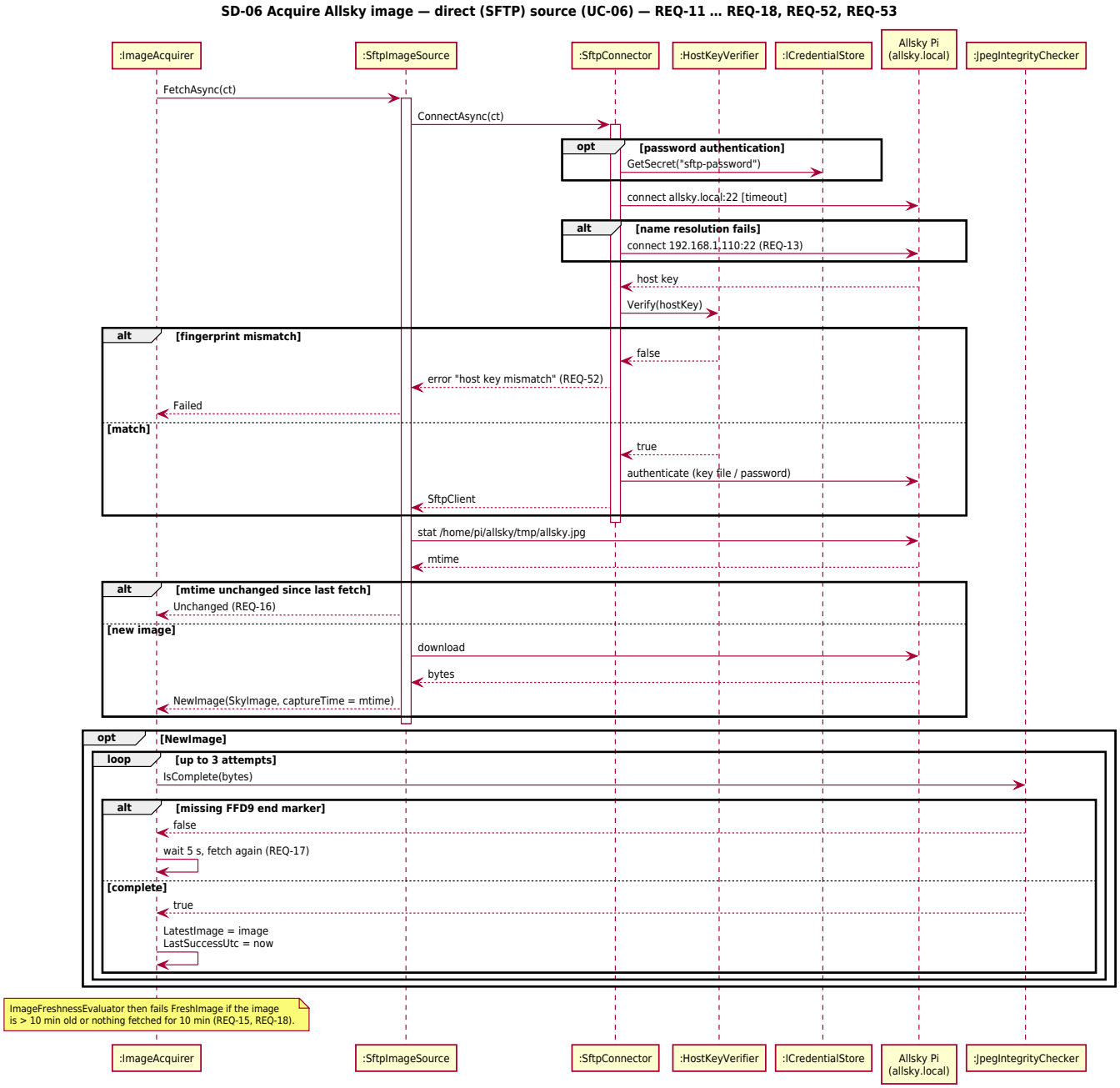


Figure 5-7 SD-07 Monitor failure and idle window

An exception or hang becomes an Unsafe status without reaching NINA. Outside the operating window the cycle idles and reports unsafe.

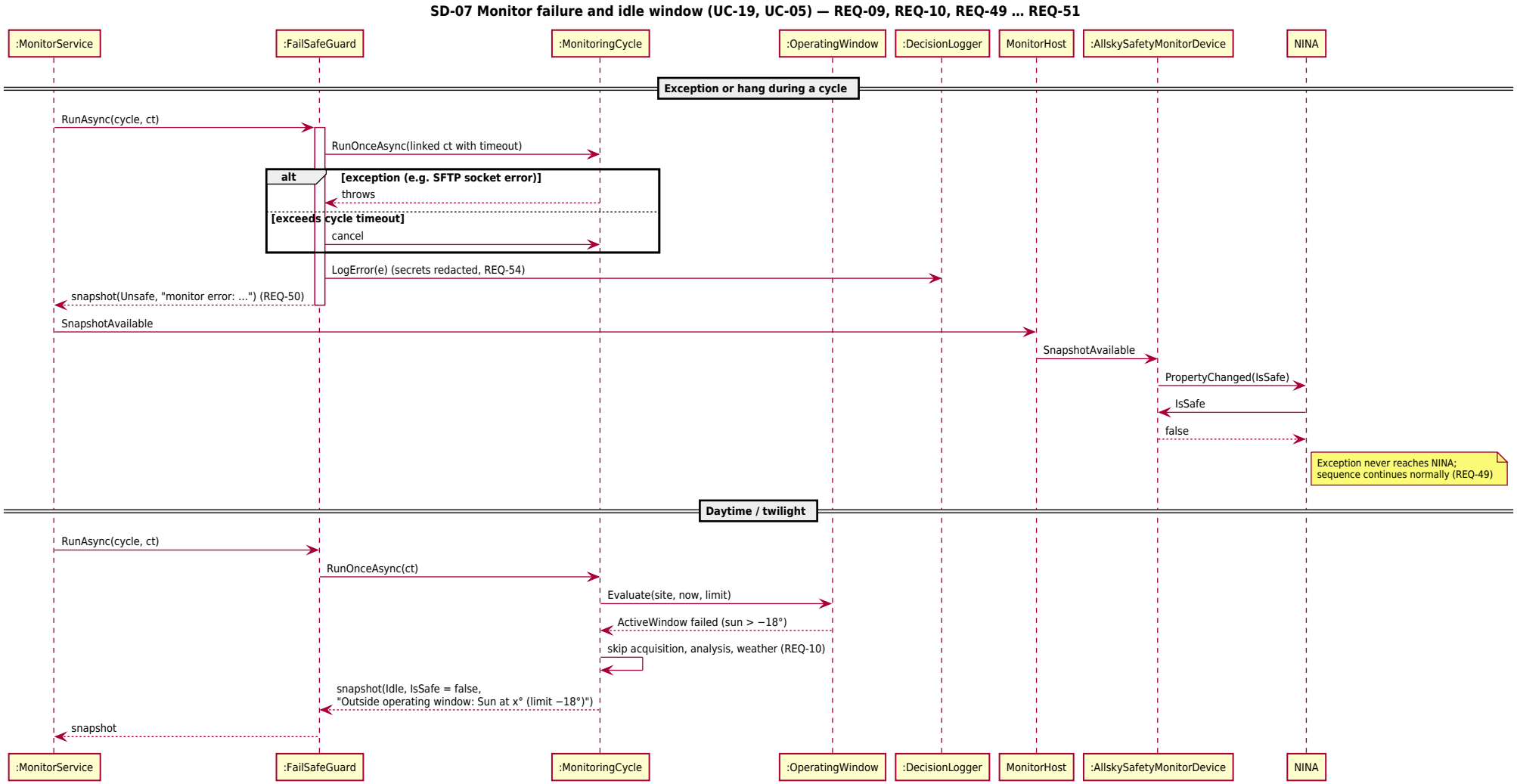


Figure 5-8 SD-08 Calibrate lens and define horizon mask

Detect, identify and fit stars, inspect the overlay and RMS error, draw the mask, accept and save to the JSON settings.

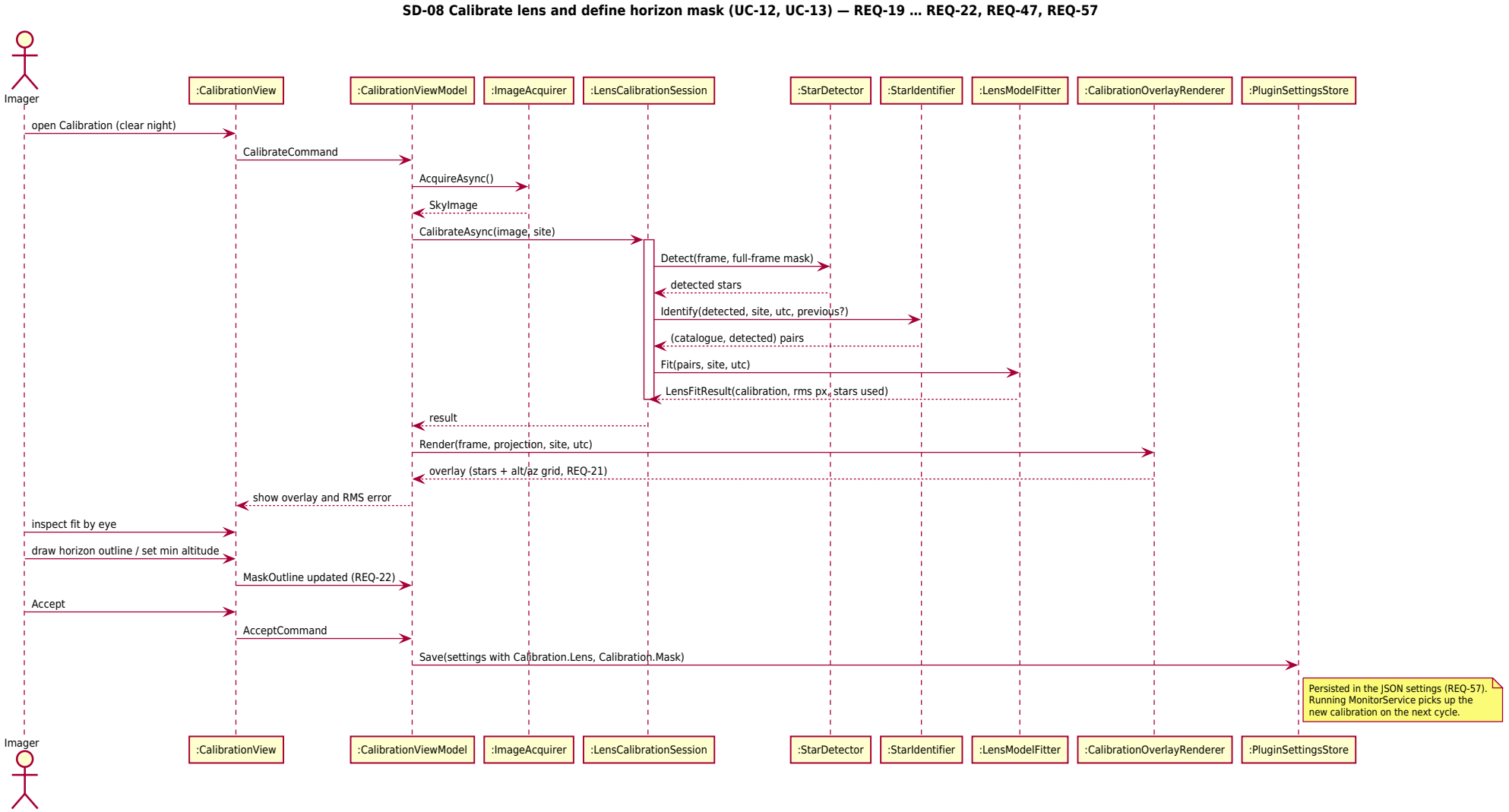


Figure 5-9 SD-09 Replay saved frames offline

Saved frames run through the real cycle with trial settings and a clock set from each frame’s capture time; results go to a CSV for Excel.

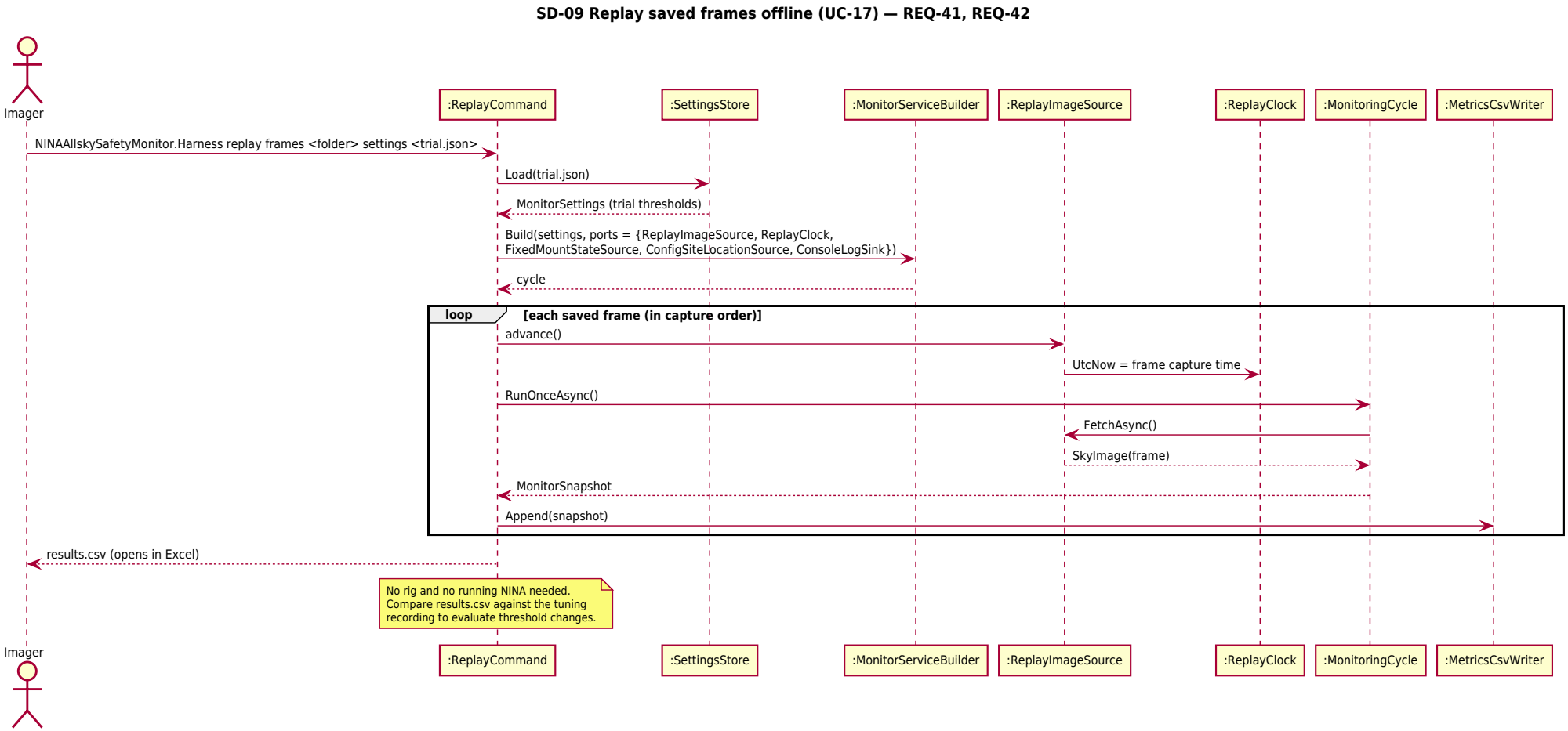


Figure 5-10 SD-10 Manual override and disconnect

Override for testing, not persisted. Disconnect stops the loop and unregisters from NINA, leaving nothing running.

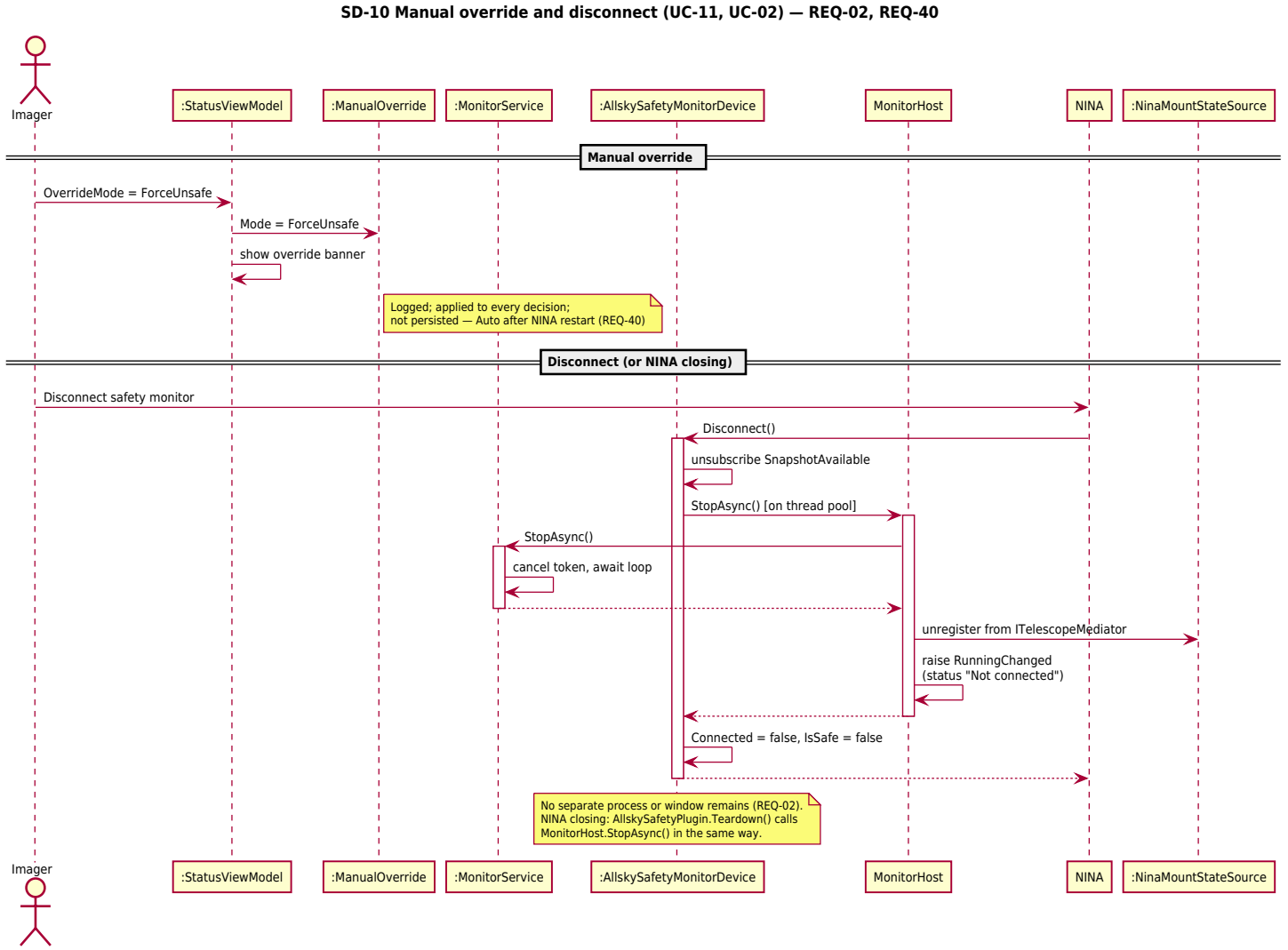
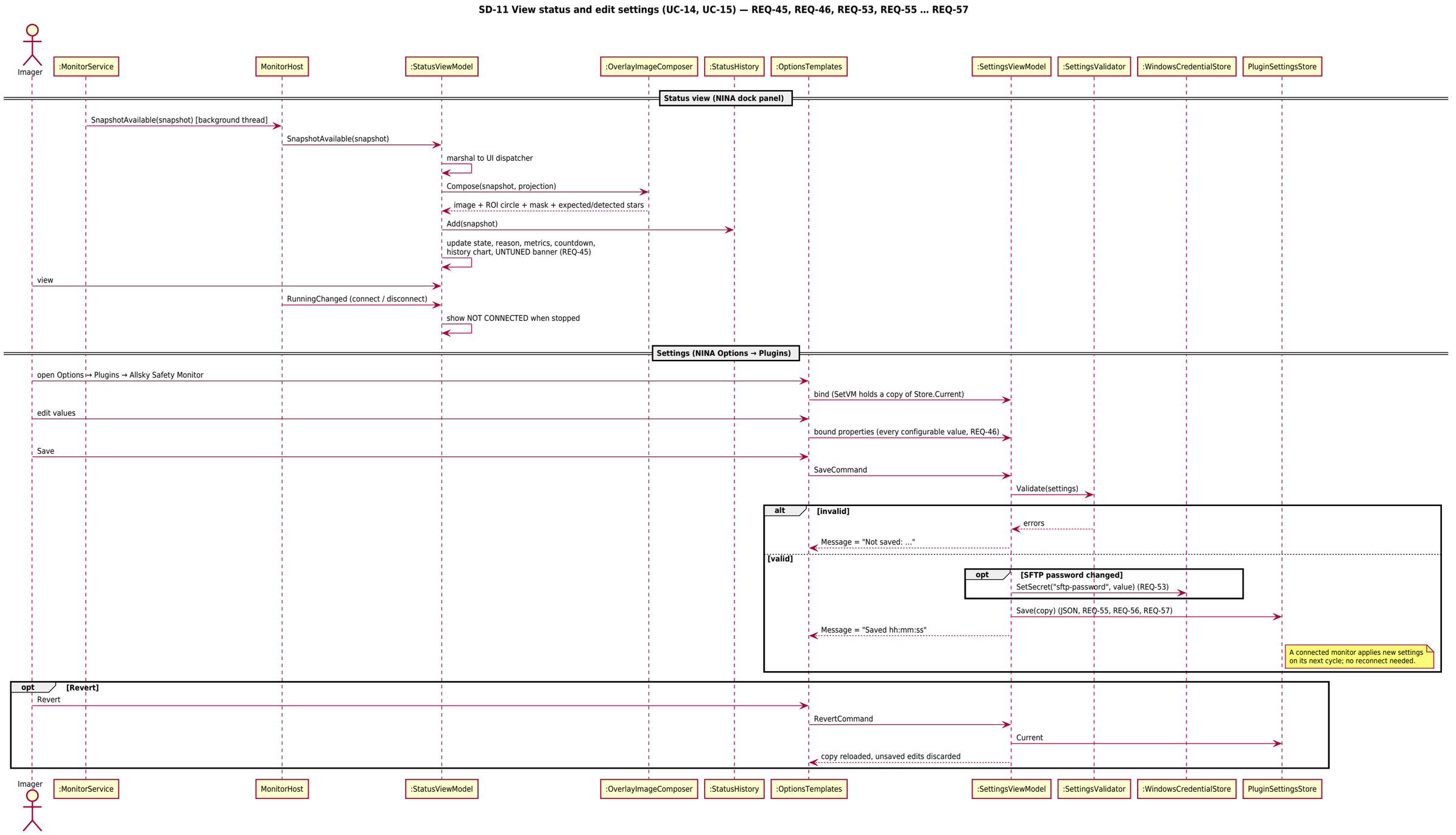


Figure 5-11 SD-11 View status and edit settings

The status panel updates from each snapshot. Settings are validated; the password goes to Credential Manager and everything else to JSON.



6 Traceability matrix

All 57 requirements trace to at least one use case, one design class and one scenario. The matrix is generated from the diagram sources; traceability.xlsx holds the same data with filters.

REQ	Requirement (abridged)	Use cases	Design classes	Scenarios
REQ-01	Appears in NINA as a safety monitor device (Equipment → Safety Monitor), usable by NINA sequenc…	UC-01, UC-02	AllskySafetyMonitorDevice, AllskySafetyMonitorProvider	SD-01
REQ-02	Starts working when the device is connected in NINA and stops when it is disconnected or NINA c…	UC-01, UC-02	AllskySafetyMonitorDevice, MonitorHost, MonitorService	SD-01, SD-10
REQ-03	Reports the result of the safety decision (§9) to NINA as safe or unsafe.	UC-03	AllskySafetyMonitorDevice, MonitorHost, SafetyDecisionEngine, SafetyStatus	SD-04
REQ-04	The site location is taken from the active NINA profile. If that isn't available, a configured …	UC-05, UC-07	ISiteLocationSource, NinaSiteLocationSource, SiteLocation, SiteLocationResolver	SD-02
REQ-05	The mount state is taken from NINA: connected, parked, tracking, slewing, RA (hours), Dec (degr…	UC-07	IMountStateSource, MountState, NinaMountStateSource	SD-05
REQ-06	The mount-reported Altitude/Azimuth are not used for positioning. The ASI Mount driver reports …	UC-07	DecisionLogger, MountState, RoiCalculator	SD-02
REQ-07	When the mount reports no valid position (for example, disconnected), the position is treated a…	UC-07	MountState, MountStateInterpreter, NinaMountStateSource	SD-05
REQ-08	Computes sun altitude from the site location and current time.	UC-05	IAstroCalculator, NovasAstroCalculator, OperatingWindow	SD-02
REQ-09	Active only while the sun is below (astronomical twilight).	UC-05	OperatingWindow	SD-07
REQ-10	Outside the active window it idles: no image fetching or analysis, and it reports unsafe.	UC-05	MonitoringCycle, OperatingWindow	SD-07
REQ-11	The image source is selectable:	UC-06	ImageSourceFactory, SftpImageSource, SkyImage, WebImageSource	SD-06
REQ-12	Direct access authenticates with an SSH key (preferred,) or a password.	UC-06	SftpConnector	SD-06
REQ-13	The Pi host is, with an optional fallback IP in case name resolution fails.	UC-06	SftpConnector	SD-06
REQ-14	Fetches a new image every while active.	UC-06	ImageAcquirer	SD-06
REQ-15	Determines the image capture time (file modification time, or the web server's last-modified ti…	UC-06	ImageFreshnessEvaluator, SftpImageSource, SkyImage, WebImageSource	SD-02, SD-06
REQ-16	For direct access, the file's modification time is checked before downloading, and the image is…	UC-06	SftpImageSource	SD-06

REQ	Requirement (abridged)	Use cases	Design classes	Scenarios
REQ-17	An image caught part-way through being written (an incomplete JPEG) is discarded and fetched ag…	UC-06	ImageAcquirer, JpegIntegrityChecker	SD-06
REQ-18	If no image can be fetched for, reports unsafe.	UC-06	ImageFreshnessEvaluator	SD-06
REQ-19	Provides a one-off calibration that maps alt/az to Allsky image pixel positions: image centre, …	UC-12	CalibrationViewModel, LensCalibration, LensCalibrationSession, LensModelFitter, LensProjection	SD-08
REQ-20	Calibration is done by matching identified catalogue stars in a real clear-sky frame.	UC-12	LensCalibrationSession, LensModelFitter, StarIdentifier	SD-08
REQ-21	Shows a verification overlay (catalogue stars and an alt/az grid drawn on the live image) so ca…	UC-12	CalibrationOverlayRenderer, CalibrationViewModel	SD-08
REQ-22	Supports a horizon mask: either a minimum altitude or a drawn mask outline, to exclude trees, t…	UC-13	CalibrationViewModel, HorizonMask, HorizonMaskDefinition	SD-08
REQ-23	The ROI is a circle of radius on the sky, centred on the mount's current RA/Dec.	UC-07	RegionOfInterest, RoiCalculator, RoiMaskBuilder	SD-02
REQ-24	RA/Dec is converted to alt/az using the coordinate epoch (JNOW on this rig), the site location …	UC-07	IAstroCalculator, LensProjection, NovasAstroCalculator, RoiCalculator	SD-02
REQ-25	Any part of the ROI below the horizon mask (REQ-22) is excluded from analysis.	UC-07	HorizonMask, RoiMaskBuilder	SD-02
REQ-26	The ROI falls back to a circle centred on the zenith when the mount is not connected, is parked…	UC-07	MountStateInterpreter, RegionOfInterest, RoiCalculator	SD-05
REQ-27	While the mount is slewing, the current safety state is held and the ROI is updated once the sl…	UC-07	SlewHoldPolicy	SD-05
REQ-28	Any cloud in the ROI means unsafe.	UC-08	CloudAssessment, CloudDetector	SD-03
REQ-29	Primary test: predict which catalogue stars should be visible in the ROI (down to a limiting ma…	UC-08	ExpectedStarPredictor, StarCatalogue, StarFractionTest	SD-03
REQ-30	Secondary test, comparing the ROI with a clear-sky reference captured automatically on the firs…	UC-08	BrightnessTest, ClearSkyReference, ClearSkyReferenceBuilder	SD-03
REQ-31	Stars within of the Moon are excluded from the expected list, and brightness thresholds are adj…	UC-08	ExpectedStarPredictor, IAstroCalculator, MoonBrightnessAdjuster, NovasAstroCalculator	SD-03
REQ-32	Polls the Weather.com PWS current-observations API for station every, with metric units, authen…	UC-09	PwsWeatherClient, WeatherPoller	SD-02
REQ-33	Values are read from observations: epoch (observation time), metric.precipRate (mm/h), metric.w…	UC-09	PwsObservationParser, WeatherObservation	SD-02
REQ-34	Any precipitation rate above zero → unsafe immediately.	UC-09	WeatherEvaluator	SD-02
REQ-35	Wind gust above → unsafe.	UC-09	WeatherEvaluator	SD-02

REQ	Requirement (abridged)	Use cases	Design classes	Scenarios
REQ-36	If the station data is older than or unavailable → unsafe. Weather checking can be switched off…	UC-09	WeatherEvaluator	SD-02
REQ-37	The overall status is safe only when all of these are true: active window, fresh image, no clou…	UC-10	MonitoringCycle, SafetyConditions, SafetyConditionsBuilder, SafetyDecisionEngine	SD-02, SD-04
REQ-38	Transition to unsafe is immediate when any condition fails.	UC-10	SafetyHysteresis	SD-04
REQ-39	Transition to safe happens only after all conditions have held continuously for.	UC-10	SafetyHysteresis, SafetyStatus	SD-04
REQ-40	A manual override (Force Safe / Force Unsafe / Auto) is provided for testing, shown prominently…	UC-11	ManualOverride, SafetyStatus, StatusViewModel	SD-10
REQ-41	A tuning mode saves frames together with their computed metrics, in a format readable by Excel,…	UC-16	MetricsCsvWriter, MonitorSnapshot, TuningRecorder	SD-02, SD-09
REQ-42	Saved frames can be re-analysed offline, without the rig or a live NINA session, to test thresh…	UC-17	FixedMountStateSource, ReplayClock, ReplayCommand, ReplayImageSource	SD-09
REQ-43	Until the thresholds have been tuned, conservative defaults (biased towards unsafe) are used an…	UC-16	SafetyDecisionEngine, SafetyStatus, StatusViewModel, TuningStateEvaluator	SD-04
REQ-44	Keeps a rolling log of state changes, metrics, errors and the reason for every safe/unsafe deci…	UC-18	DecisionLogger, RollingFileLogSink	SD-02
REQ-45	A status view shows:	UC-14	HistoryChartViewModel, MetricsViewModel, MonitorSnapshot, OverlayImageComposer, StatusHistory, StatusViewModel	SD-11
REQ-46	A settings view edits every configurable value.	UC-15	AllskySafetyPlugin, SettingsViewModel	SD-11
REQ-47	A calibration view supports lens calibration (REQ-19–21) and drawing the horizon mask (REQ-22).	UC-12, UC-13	CalibrationViewModel	SD-08
REQ-48	Runs on the Windows AstroPC within NINA 3.x (currently the 3.3 nightly).	UC-01	AllskySafetyPlugin	SD-01
REQ-49	A failure in the safety monitor (crash, hang, network fault) must not disrupt NINA or a running…	UC-19	AllskySafetyMonitorDevice, FailSafeGuard, MonitorService	SD-07
REQ-50	Fail-safe: any failure, or any state that can't be determined, results in unsafe.	UC-19	FailSafeGuard	SD-07
REQ-51	Every network operation (image fetch, weather poll) has a timeout, so a stalled connection can'…	UC-19	FailSafeGuard, ImageAcquirer, PwsWeatherClient	SD-07
REQ-52	The Pi's host key is verified against a pinned fingerprint; a mismatch is reported and the imag…	UC-06	HostKeyVerifier	SD-06
REQ-53	An SFTP password, if used, is stored using Windows' credential storage, never in the settings f…	UC-06, UC-15	ICredentialStore, WindowsCredentialStore	SD-06, SD-11

REQ	Requirement (abridged)	Use cases	Design classes	Scenarios
REQ-54	Passwords and API keys are never written to the log.	UC-18	SecretRedactor	SD-07
REQ-55	All configurable values persist between sessions.	UC-15	MonitorSettings, PluginSettingsStore, SettingsStore, SettingsViewModel	SD-01, SD-11
REQ-56	Settings are stored in a JSON file.	UC-15	PluginSettingsStore, SettingsStore	SD-01, SD-11
REQ-57	Calibration results (REQ-19), the horizon mask (REQ-22) and the clear-sky reference (REQ-30) ar…	UC-15	CalibrationData, ClearSkyReference	SD-08, SD-11